

An aerial photograph of a city grid, likely Melbourne, Australia, with a blue overlay. A dashed white line runs diagonally from the top left towards the bottom right. The word 'PRECINCT' is written in a large, white, outlined font at the top left.

PRECINCT



INDUSTRY INTERCONNECTORS

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June 2024

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ACKNOWLEDGEMENT OF COUNTRY

We would first like to acknowledge the traditional custodians of the land. We work on the land of the Bunnrong people in Caulfield and our project site, Sunshine North is situated on land of the Wurundjeri Woi Wurrung people of the Kulin Nation. We would like to pay our respects to the elders past and present.



A collaborative work by 22 senior First Nations women capturing the nature of movement and the connection to country, available for viewing at the National Gallery of Victoria.

Kanaputa, 2010 (image source: National Gallery of Victoria)



**WURUNDJERI WOI
WURRUNG**

SUNSHINE NORTH

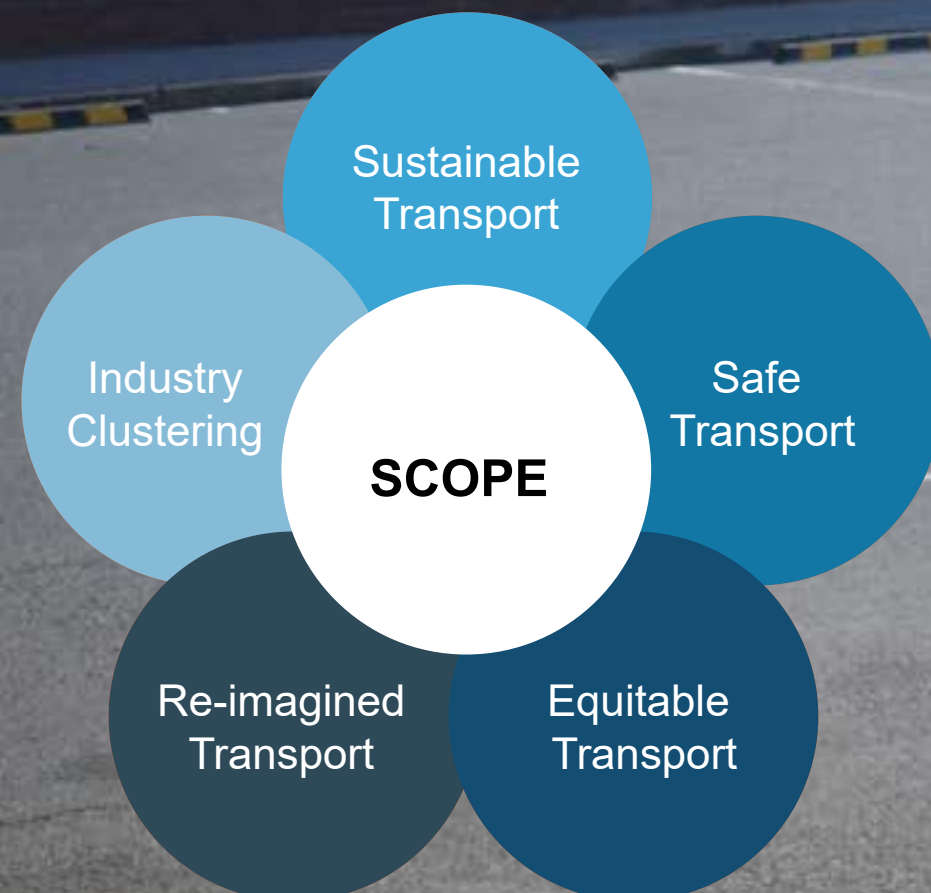
WADAWURRUNG

BUNURONG

1. INTRODUCTION

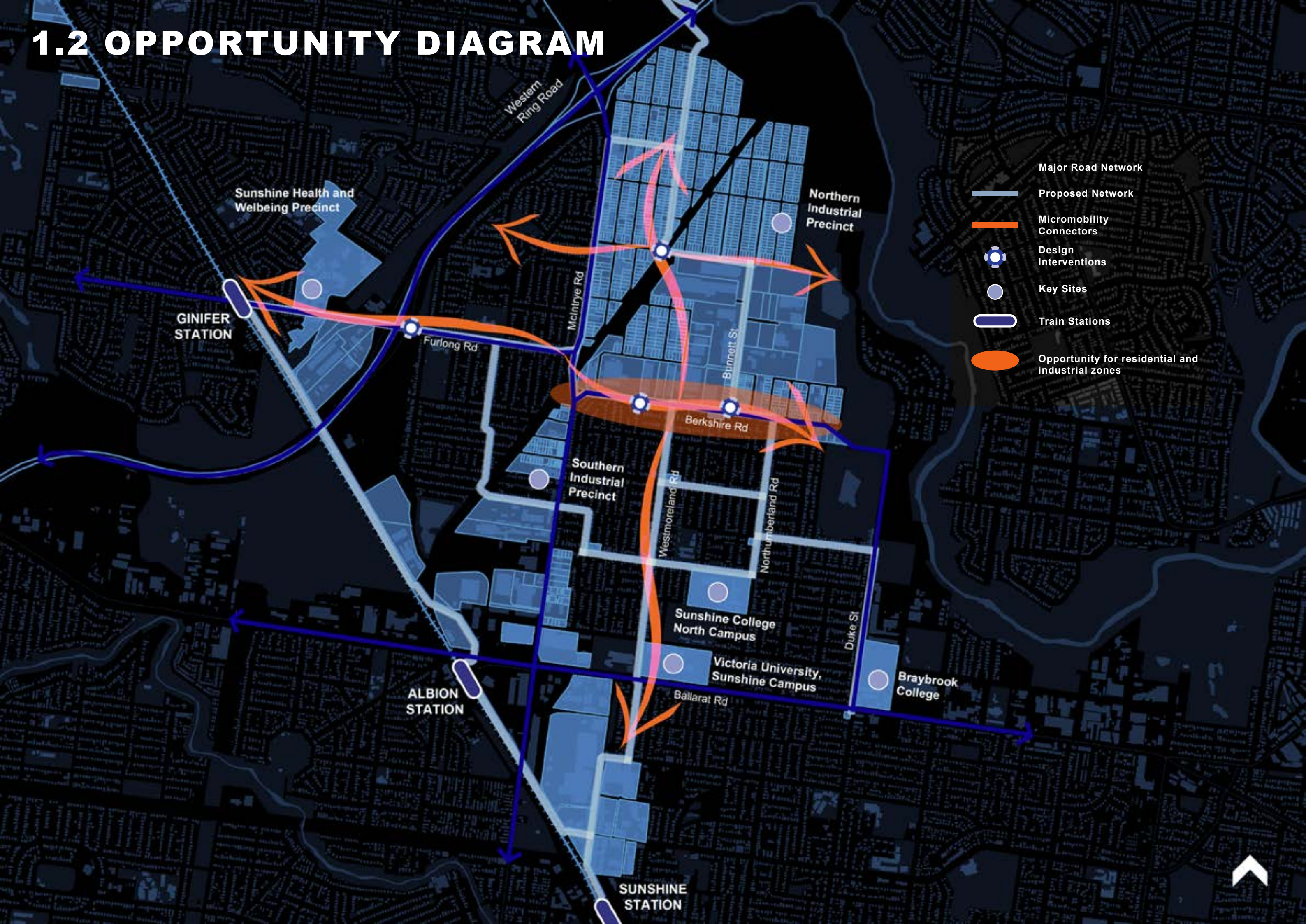
Our vision for Sunshine North is to interconnect critical industry clusters together through reimagined transport futures.

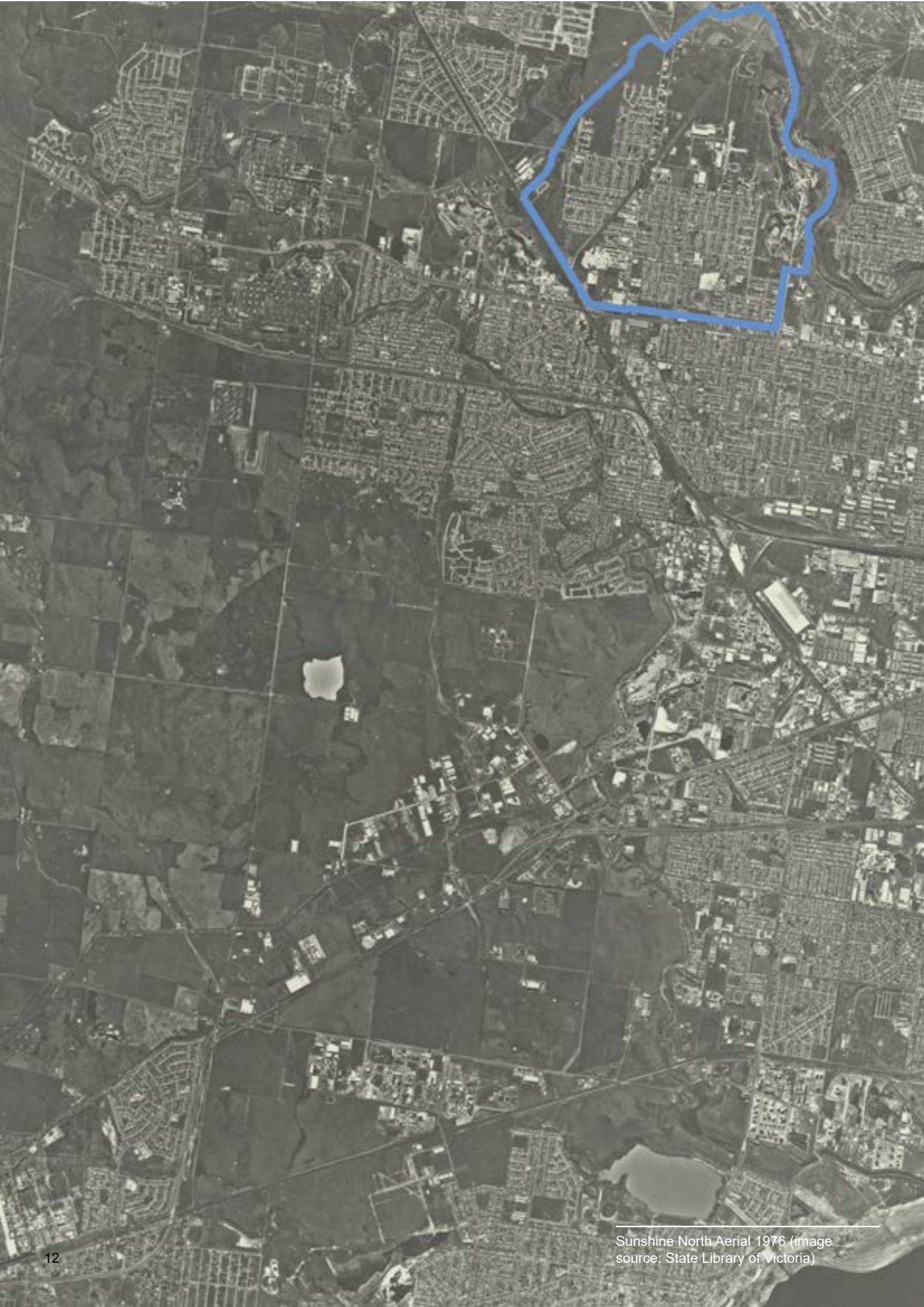
We seek to build on strong existing foundations to drive place-based development prioritizing the residents of Sunshine North.



Mighty Commodore at Dan Murphy's
(Photo credit: Rifqi Rafiq)

1.2 OPPORTUNITY DIAGRAM





Sunshine North Aerial 1976 (image source: State Library of Victoria)

Sunshine North

Sunshine North presents an opportunity pioneer future modes of industry connection. With ecological edges, rail connections, invaluable pre-existing industrial land and a plethora connection to central Melbourne and regional Victoria, Sunshine North has the properties of a burgeoning hub where industry and amenity can meet. We foresee a network connecting the suburb together which prioritizes transit for micromobility modes and active transport. Bicycles, trikes, e-cargo bicycles, small delivery vehicles and the like will flourish in an environment that keeps them safe and efficient.

Setting up a physical network will be just as important as underpinning policies guiding the modal transition of the area. We believe that by investing in both the physical and the political the industry of Sunshine North will begin to take on a new shape. One that is greener, safer, and more productive.



SUNBURY
27KM



SUNSHINE NORTH



MELBOURNE CBD
13KM



2. FRAMING THE TRANSPORT ISSUE



Sunshine North River Valley Project
(image source: Anagha Khude)

Historical Context

Sunshine Rising

Sunshine emerged as a significant town in the late 19th and early 20th century as it was the junction of the railroads that joined Melbourne and ports in Williamstown & Newport to Ballarat and Adelaide in the west and Bendigo to the north.

This propegated emergent industries whose product could be exported to most parts of Victoria and South Australia or exported overseas.The dominant industry during this time was H.V.McKay's wheat harvester manufacturing plant.

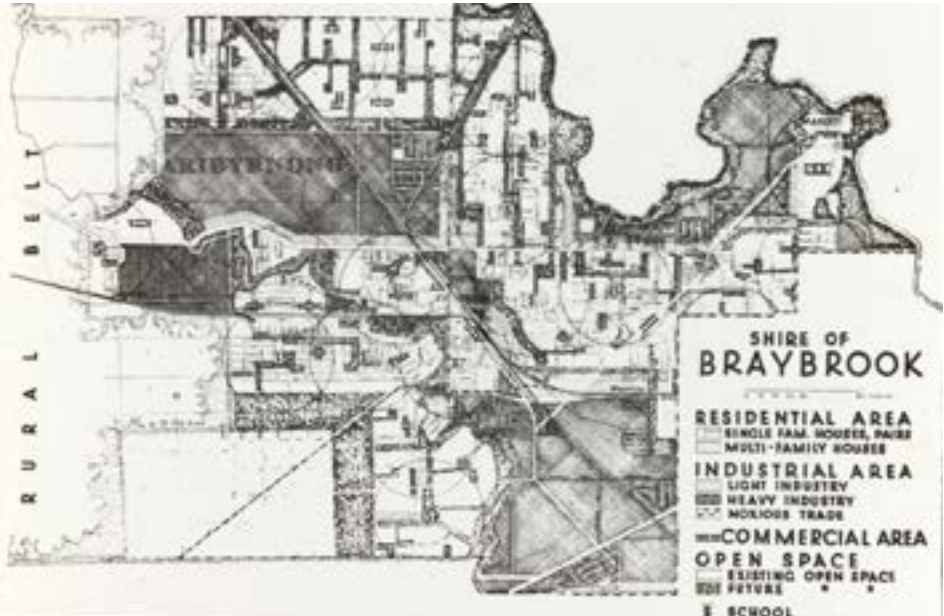
After WWI, returned soldiers settled in the newer estates creating demand for more amenities.

After the second World War, immigrants from regions suffering upheaval migrated to Australia in several waves, with many settling in newer estates in the City of Sunshine, creating a multicultural society such that by 2001, 40% of residents were from non-English speaking background (Ford. 2001).



Hampshire Road,
Sunshine

Hampshire Road Looking
North (photo credit: Massey
Ferguson, 1955)



Land use map of the
shire of Braybrook,
1976

Shire of Braybrook (image
source: State Library of
Victoria)

INDUSTRY ...

Sunshine +
Sunshine North

Footscray and Braybrook are two highly-developed commercial and industrial centres. Their well-planned cities, permanently-constructed roads and excellent shopping centres play an integral part in the general scheme. "Forging Ahead" has set out to illustrate this fact. It is a link with the general policy that has given this western metropolis the distinction of being termed the "Birmingham of Australia." It could be safely claimed that every type and variety of industry and manufacture is represented within the Footscray-Braybrook boundary.

The industrial undertakings described in this booklet do not form a complete coverage, but they are truly representative of the modern type of factory that exists and of up-to-date lay-out.

Vision in the past has brought this about, and vision at present operating, with the invaluable aid of scientific planning, makes the potentialities unlimited.

There is a grand goodwill basis between employer-employee . . . the family man is encouraged and assisted to own his own home . . . teams from various factories compete in organised sport, with the chief of the firm invariably as an active president . . . there is the fullest co-operation in anything that aims to make for a fully-occupied, happy community.

Footscray is 3½ miles from Melbourne, and the boundary of Braybrook is within four miles of Melbourne. Both places have the advantage of a fast electric train service, and modern main roads serve the areas from all points of the compass. The Victoria Docks (Yarra River) are even closer than the City, being on the route, and it is said the road between Braybrook and the docks carries more tonnage than any other road in the world. This gives an insight into the strength of the industrial undertakings, but perhaps the appended figures, taken from 1945-1946 statistics, provide a more factual indication of the colossal nature of the activities:

For the period, Footscray employed 15,465 persons and Braybrook 11,193; salaries and wages paid were £4,711,099 and £3,728,543 respectively, and the value of articles produced was £23,926,766 (Footscray) and £13,409,737 (Braybrook). There were 245 factories in Footscray and 83 in Braybrook.

Page 20 from Shire of
Braybrooks "Forging Ahead"
plan, 1947. (Image source:
State Library of Victoria)

Early Industry Development

During the Nineties there was a redevelopment process in the locales industry. Older sites housing HV Mackay, Spalding, Nettlefolds were transitioned to areas for retail whilst an emergent Sunshine Hospital sprung up in the now called Sunshine Health and Wellbeing Precinct (SHWEP).



An aerial view of the Sunshine Harvester Works circa 1950.

Aerial view of Sunshine Harvester Works, Victoria (Photo credit: Charles Daniel Pratt (1950-1954).



ETA Factory, Sunshine 1960

ETA Factory, Sunshine 1960 (photo credit: Sievers Wolfgang)



John Lysaght Australia premises at Sunshine 1966

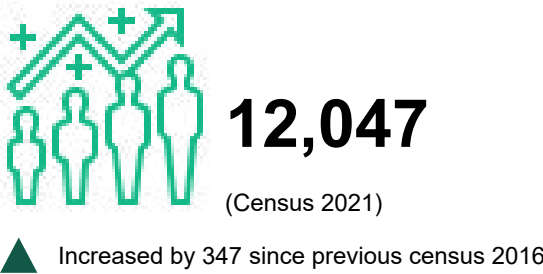
John Lysaght Australia premises, 1966 (photo credit: Sievers Wolfgang)

The old Sunshine Highschool became a center for retail and tertiary education whilst older factory sites toward the Maribyrnong becomes residential.

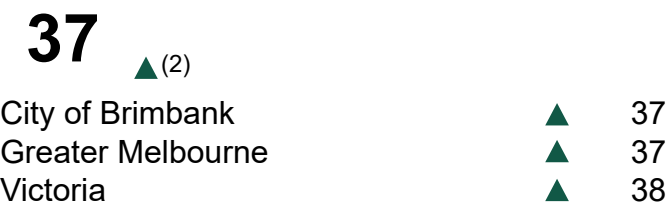
The opening of the Western Ring Road to the north of Sunshine North brings about a new composition of transit making the north-south arterial, McIntyre Road a thoroughfare of freight and traffic (Ford 2001).

Who is Sunshine North?

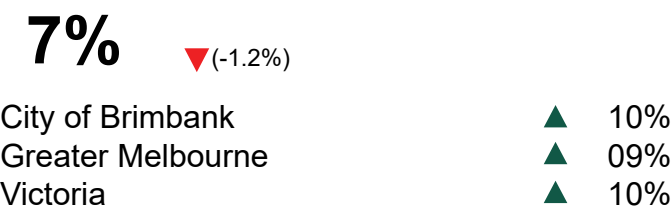
Usual resident population for Sunshine



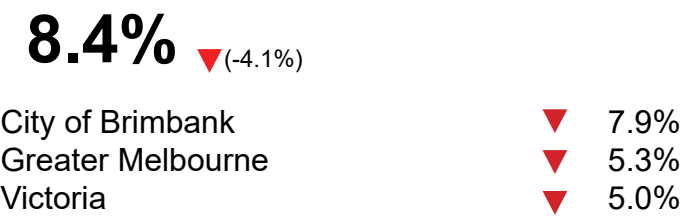
Median Age



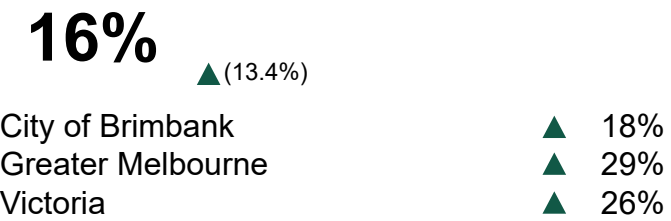
Older couples without children



Unemployment rate



Worked at home



G.K.N. warehouse , Sunshine 1969

G.K.N. warehouse Sunshine 1969 (photo credit: Sievers Wolfgang)

As per the 2021 Australian Census, Sunshine North is now comprised of approximately 12,047 people. The median age remains close to the state average at 37. With property prices in similarly distanced suburbs sky rocketing and Brimbank (LGA) retaining a shortfall in population post-covid (SGS Economics, 2024), Sunshine North may begin to appeal to younger families looking to enter into home-ownership.

The unemployment rate in the Sunshine North was 8.4% in 2021, hovering well above the state average. It is important to note the impact of the 2021 census being the 'COVID census' and thus encompassing demographic anomalies.

16% of Sunshine North worked from home in 2021., Despite this being the COVID census it is important to note that the increase from the previous census (13.4%) may be one indicator of a transforming economy in Brimbank - moving toward a knowledge economy (State Government of Victoria, 2017)

▲ No significant change since previous Census (less than +/-0.5%) ▼ Decreased since previous Census ▲ Increased since previous Census



6,051



5,990

Couples with children

33% ▼ (-2%)

City of Brimbank	▼	35%
Greater Melbourne	▼	35%
Victoria	▼	31%

Lone persons households

21% ▲ (1.5%)

City of Brimbank	▲	19%
Greater Melbourne	▲	24%
Victoria	▲	25%

Sunshine North had seen a decrease in couples with children following the state wide trend. Congruently, household compositions saw lone person households increase in stead with the rest of the State.

Participation rate (population in labour force)

54% ▲ (1.6%)

City of Brimbank	◀▶	56%
Greater Melbourne	▲	64%
Victoria	▲	62%



Lai Bakery, Sunshine (photo credit: R. Rafiq)



Lai Bakery 2, Sunshine (photo credit: R. Rafiq)

◀▶ No significant change since previous Census (less than +/-0.5%) ▼ Decreased since previous Census ▲ Increased since previous Census

Socio-Economics of Sunshine North

SEIFA INDEX Disadvantage 2021

865

City of Brimbank	▲	912
Greater Melbourne	▲	1018
Victoria	▲	1010

The Social Economic Index for Areas is a measure composed by the ABS to indicate the advantage and disadvantage of a particular geography. There are two measures of which we have used the Index of Relative Socio-Economic Disadvantage to better understand the experience of Sunshine North residents. Sunshine North's indicie of 865 indicates relative disadvantage experienced in the area scoring lower than the greater Brimbank area and the State as a whole. This indicates that on scales of education and employment, residents may have achieved lower levels of education or may be in occupations earning less and thus contributing to lower purchasing power. Additionally a lower score indicates a lesser opportunity to access economic resources like public services, education and additional care¹.



Local street art on rail controller box

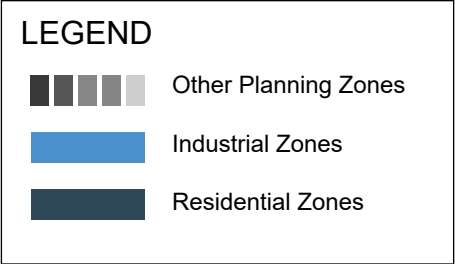
Jacana Railway, Sunshine North (photo credit: R. Rafiq)

◀▶ No significant change since previous Census (less than +/-0.5%) ▼ Decreased since previous Census ▲ Increased since previous Census

Strong Edges, Strong Foundation



We are able to distinguish between the residential and industrial/commercial quarters (left). With the Maribyrnong to the west, the M80 ring road to the North, the dissecting Jacana-Albion railway through the centre and Ballarat Road in the South there are very strong spatial edges in Sunshine North.

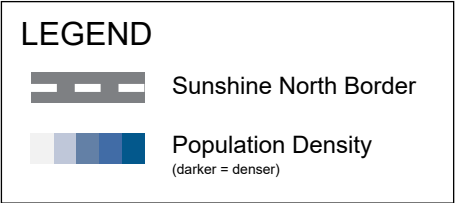


Land use separation in Sunshine North

Sunshine North Zoning Map (image source: R.Rafiq)



Further we are able to see the arrangement of residential density within those quarters. Built up areas of residence toward the periphery and centreline of the residential quarters. Furthermore there is a good deal of density in and around the locale. This reminds us that Sunshine North does not operate in a vacuum but in fact is part of a larger spatial ecosystem. With the strong transit paths (M80, McIntyre Road, Sunbury Rail etc.) Sunshine North experiences a wealth of pass-through activity.



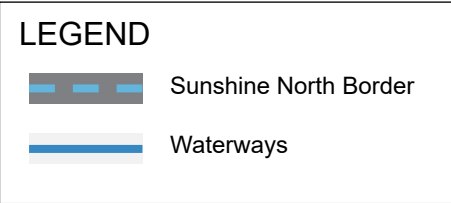
Density illustrated across Sunshine North.

Sunshine North Density Map (image source: R. Rafiq)



Compounding the notion of strong edges the Maribyrnong makes up only a portion of the natural features in Sunshine North. With confluencing Stony and Kororoit Creeks toward the Western part of Sunshine North and the further North Jones Creek, Sunshine North boasts opportunities to sensitively interact and develop with these ecological systems. It is important to note a previous less-sensitive interaction where-by the Stony Creek has been redirected to avoid the developing Albion Quarter.

Lastly it is important to consider that the residential developments toward the Maribyrnong lay proximal to the 100 year flood zones as applied by the Department of Transport and Planning.



Waterways in Sunshine North along with contours

Sunshine North Water and Contour Map (image source: R. Rafiq)

Sunshine North: Location of Precincts and Edges



LEGEND

- Key Precinct
- Precinct Built Environment
- ← - - - Precinct Pathways

Sunshine Edges and precinct Location (image source: R. Rafiq)

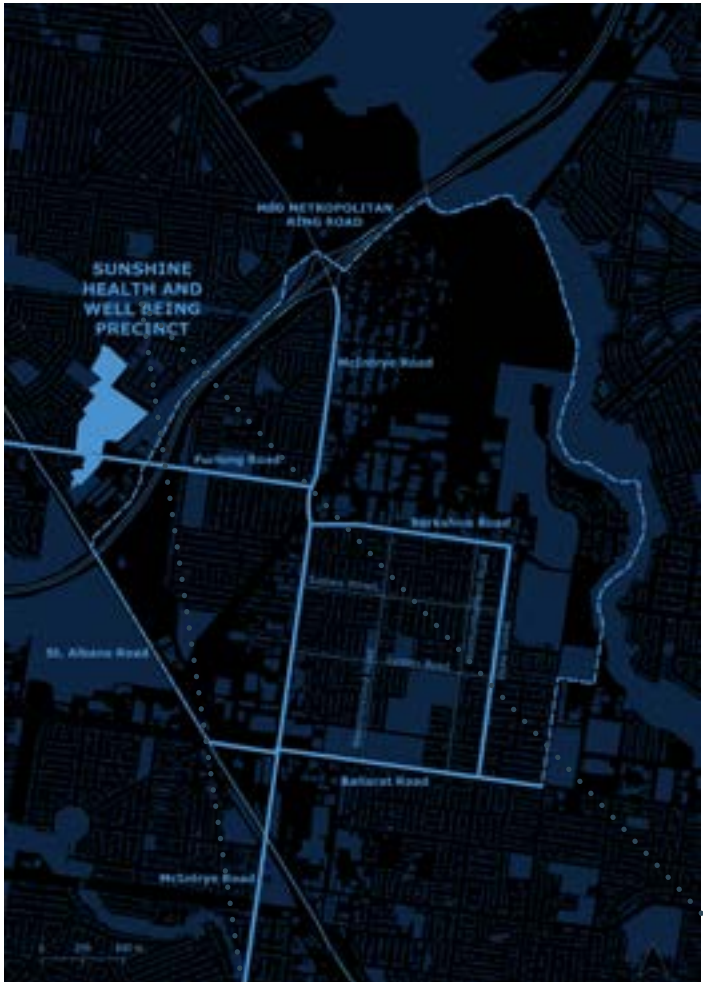
When considering our site it was important to spatialise the major nodes, their compositions and the current pathways that connect them. Our key precincts are:

- 1. Sunshine Health and Wellbeing Precinct
- 2. Northern Industrial Precinct
- 3. Education Precinct
- 4. Southern Industrial Precinct

Connecting these major nodes are the:

- 1. M80 Metropolitan Ring Road
- 2. Furlong Road
- 3. Ballarat Road
- 4. McIntyre Road

1. Sunshine Health and Wellbeing Precinct



Additional Proximal Services

- Private Health Consultants
- Physiotherapists
- Fronditha age Care
- Cover Half Disability Services
- Pathology
- Research Centre
- Health Club
- Radiology, ICU, Community Health Services
- Eye Surgeons

LEGEND

- Sunshine North Border
- Major Paths
- Precinct

Sunshine Health and Wellbeing Precinct location (photo credit: R. Rafiq)



Sunshine Private Hospital (image source: BKK Architects, 2017)

2. Industrial Precincts : North + South



Industrial Precincts

The industrial precincts in Sunshine North are comprised of various industries. Of the most competitive Sunshine North has a specific specialisation in manufacturing and other services (inc. automotive services, repair, body works etc.). This speaks to the overall strength of having the built environment to accomodate warehouse and yard based industries. Manufacturing of which the Australian Government is looking to strengthen is largely based in sites akin to these (Future Made in Australia, 2024).

LEGEND

Sunshine North Border

Major Paths

Precinct

Industrial Precinct location (photo credit: R. Rafiq)

Sunshine North Industries of Competitive Advantage

- Equipment/ Material Distribution
 - Construction¹
- Automotive Repair
 - Manufacturing
 - Wholesale trade



Empty Site on McIntyre road (photo credit: R. Rafiq)



Auto Repair on Suffolk rd (photo credit: R. Rafiq)

1. Brimbank City Council, “Brimbank Economic Development Strategy”

Dominant Industries (LQ>1) ¹

A location analysis on the industries of Sunshine North revealed that the most competitive industries are manufacturing and other services. This is to say that compared to the greater Melbourne reference economy, Sunshine North’s manufacturing and other services account for a larger proportion of their overall economy. This often will be a barometer for industry clustering and agglomeration within a locality.

INDUSTRY LEVEL 1	LOCATION QUOTIENT
Mining	1.68
Manufacturing	3.51
Electricity, Gas, Water and Waste Services	1.37
Construction	1.67
Wholesale Trade	1.47
Other Services	2.51

Largest Workforce in Dominant Industries

INDUSTRY LEVEL 3	SUNSHINE WORKFORCE
Automotive Repair and Maintenance	288
Polymer Product Manufacturing	184
Hardware, Building and Garden Supplies Retailing	184
Motor Vehicle and Motor Vehicle Part Manufacturing	133
Petroleum and Coal Product Manufacturing	128
Building Completion Services	122
Basic Ferrous Metal Manufacturing	106
Building Installation Services	91
Residential Building Construction	89
Clothing and Footwear Manufacturing	87
Furniture Manufacturing	87

Taking a closer look at the dominant workforces within those industries we can see a clear emergent automotive repair and maintainence workforce in addition to plastic manufacturing, garden and hardware retail and motor vehicle related manufacturing.



Auto Repair on McIntyre rd (photo credit: R. Rafiq)

Better understanding the composition of the industrial precincts foregrounds our design considerations when looking to service the needs of these specialised industries.

1. Australian Bureau of Statistics, “Sunshine North, Industry Level 1 Data: ABS Australian Census (2021).”

3. Education Precinct



Victoria University, Sunshine College, North Campus, Braybrook College are three major higher educational institutes in and around Sunshine North.

Victoria University has two campuses in the City of Brimbank located in St Albans and in Sunshine.

LEGEND

Sunshine North Border

Major Paths

Precinct

Education Precinct location (photo credit: R. Rafiq)



Victoria University Sunshine (photo credit: Victoria University)

Victoria University, Sunshine
Courses and Schools¹:

- Clinical School of Health and Paramedicine
- TAFE: Building and Construction, Electrical Services
- Information Technology
- Engineering and Science
- Arts and Humanities

1. Victoria University, “Victoria University Sunshine Courses.”

Sunshine College, North Campus



Sunshine College North Campus. (photo credit: Sunshine College)

Braybrook College



Braybrook College Campus (photo credit: Braybrook College)

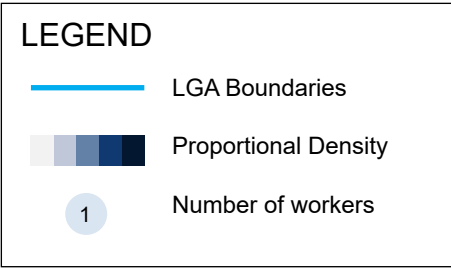
- Courses and Schools:
- VCE Courses
 - Technical Certification in Building and Construction, Engineering, Metal working
 - ATAR Programs
 - Selected Arts Programs



This map illustrates the spatial distribution of Sunshine North residents and where they live.

We can observe that the majority of the workers in 2021 made their work journey within the local of Brimbank. Proximal LGA Maribyrnong was the second highest observed destination with Melton following there after.

A major takeaway from this is the locality of many Sunshine North residents' workplace. By providing adequate infrastructure to propegate safe and environmentally conscious modes of transport, Sunshine North will have the opportunity to further build on strong foundation.

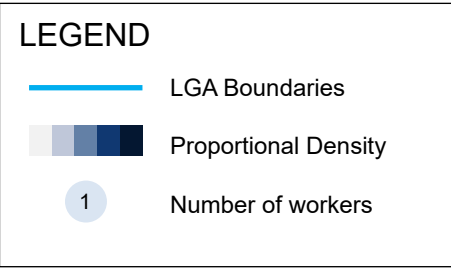


Location of Sunshine North residents workplace (LGA)

Sunshine North Residents' Workplace LGA (image source: R. Rafiq)

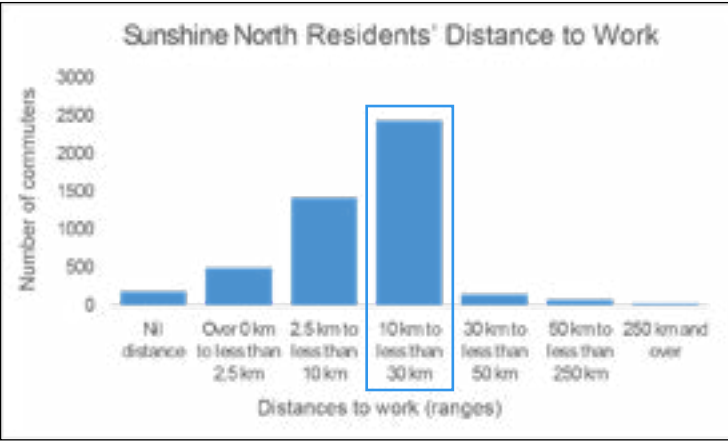


Furthermore this map illustrates the inverse. The origin LGA of the Sunshine North workers. We are able to see that although there is a concentration of Brimbank based wrkers, Sunshine is the workplace destination of many and from various localities. Melton, Hume and Wyndham City make up the larger proportions of Sunshine North workers illustrating the importance to the livelihoods of those outside of the locale.



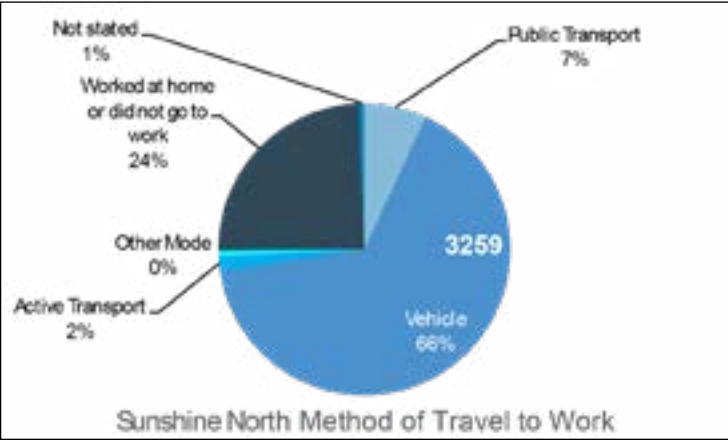
Location of Sunshine North workers place of work (LGA)

Sunshine North Workers residence LGA (image source: R. Rafiq)



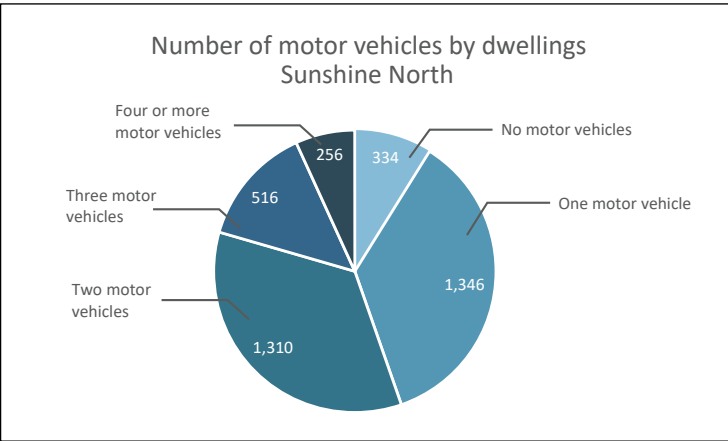
Distance to Work, Sunshine North (2021). (image source: R. Rafiq)

Compounding the Sunshine North’s Strong foundations, this graph illustrates the highest proportion of Sunshine North residents’ commute distance. We are able to observe that the largest proportion travel between 10-30kms, with the second most prominent distance range being 2.5kms-10kms.



Method of Travel to Work Sunshine North (2021) (image source: R. Rafiq)

Though we are able to observe somewhat short distances in journey to work we also observe that Sunshine North (3259 residents) surveyed 66% of the population using their car to get to work.



Motor Vehicles by Dwelling Sunshine North (2021) (image credit: A. Khude).

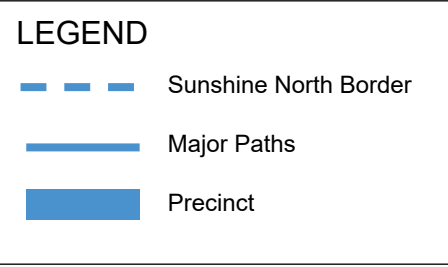
In the above illustration we are able to observe the ownership of motor vehicles per dwelling in Sunshine North of which nearly half of the population make up owners of 2 or 3 motor vehicles.

Mobility Transport Analysis

Road networks



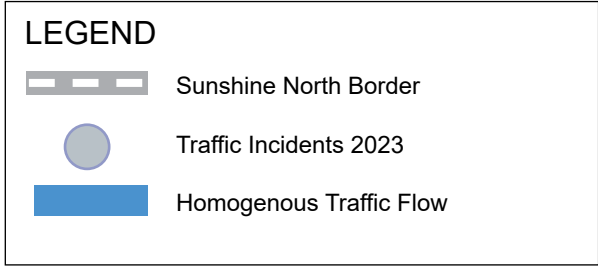
The road networks of Sunshine North are currently strong pathways for cars to travel through but unfortunately leave alternative forms of transport in unamenable and unsafe circumstances.



Major Roads in Sunshine North (photo credit: R. Rafiq)



We can observe in the map to the left that there is a pattern of crash incidents denoted by the hollow bubbles which largely coincide with the roads which see the largest volume of use. Additionally there are secondary order networks emerging on uncatagorized roads illustrating local thoroughfares chosen by drivers within Sunshine North.



Traffic Flow and Incidents, Sunshine North (photo credit: R. Rafiq)



Suffolk Road POV (photo credit: Anagha Khude)



McIntyre Road and Furlong Road intersection (photo credit: Anagha Khude)



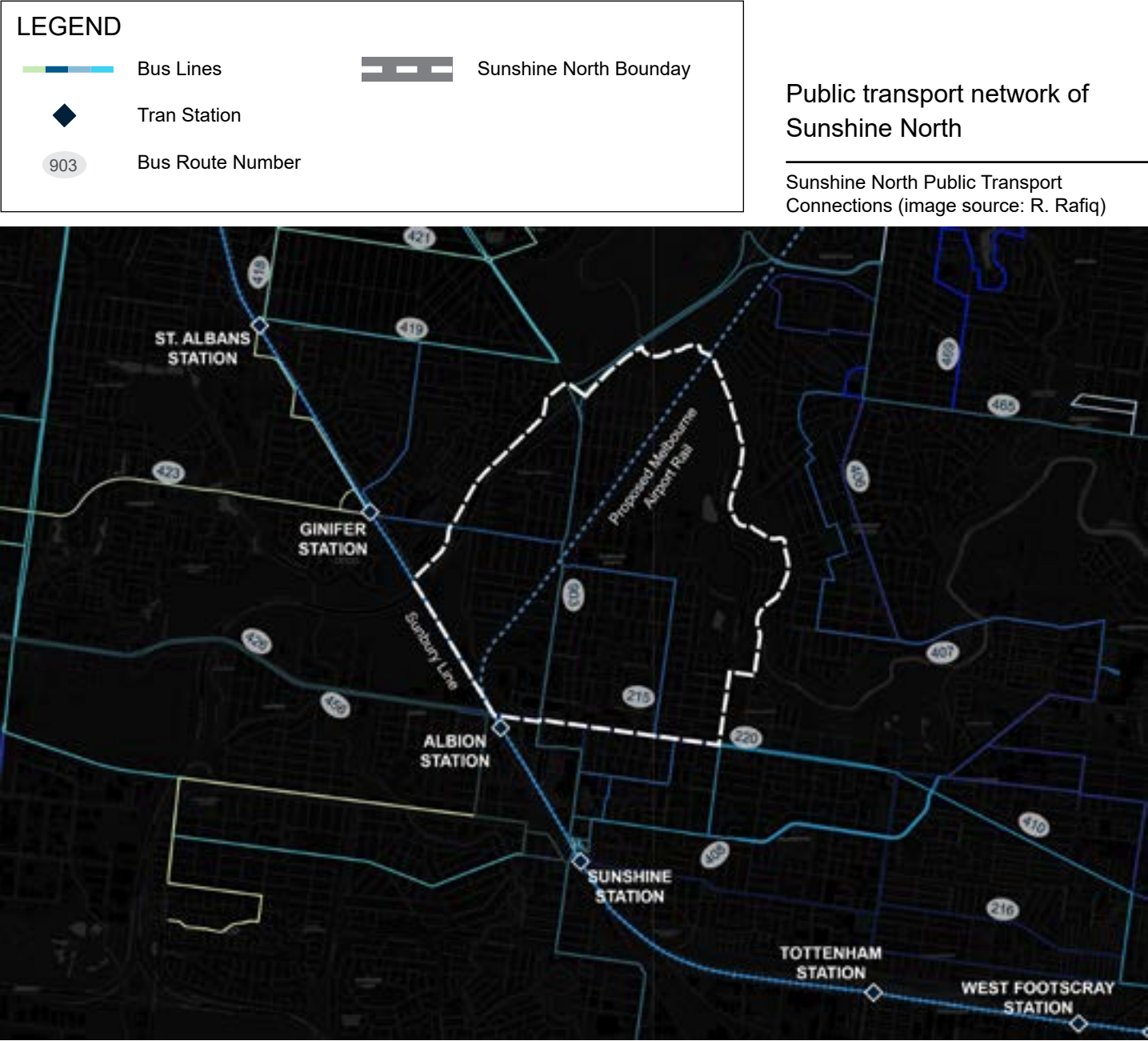
MyIntyre Road bridge (photo credit: Anagha Khude)



Furlong Road (photo credit: Anagha Khude)

Public transport network

- Sunshine North is currently serviced by one train route and various internal and peripheral bus routes.
- The Sunbury line services patronage from the Melbourne CBD out to the V Line regional connection out to Sunbury. This rail line sits on the further eastern border of Sunshine North and if well connected by a servicable bus line would facilitate an efficient alternative to cars.
 - Currently bus services within Melbourne’s West in general lack the frequency and reliability needed to gain the trust and patronage of residents. The 215 and 220 lines are infrequent Sunshine North servicing buses that at peak time will be available in 15 minute increments but falling anywhere outside of peak will leave a rider stranded for ~40 minutes. The Smart Bus route: 902 on the other hand is a prime mover of people and provides a vital connection through the heart of Sunshine North.

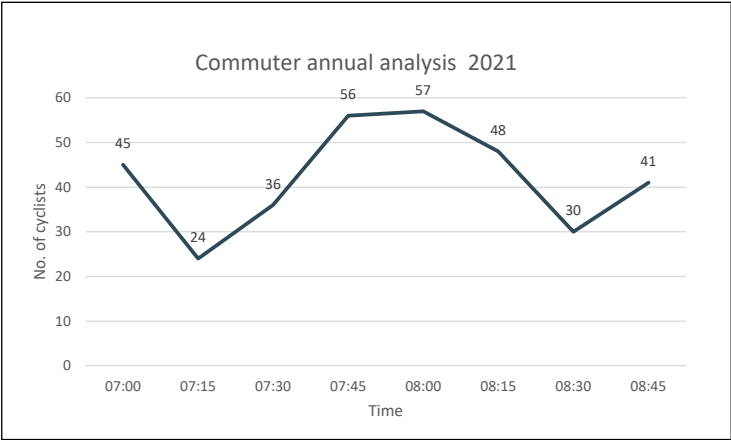


Cycling network and Open space



We can see in the above illustration a plethora of open space walking catchments, adjacent to bicycle lanes and proposed bicycle infrastructure. Unfortunately at the time the servicability of the infrastructure is leaves much to be desired for the bicycle riders of Sunshine North. the level of service on many of the roads stops at a painted line and a bicycle illustration.

Though the level of current service remains low the foundation of wide streets, a proposed network and strong links (rivers, creeks etc.) make the implementation of higher service imaginable.



Bike Commuter Annual analysis 2021 City of Brimbank (Source: Data Dashboard, Transport Surveys and Data 2021)

Qualitative Mobility Analysis

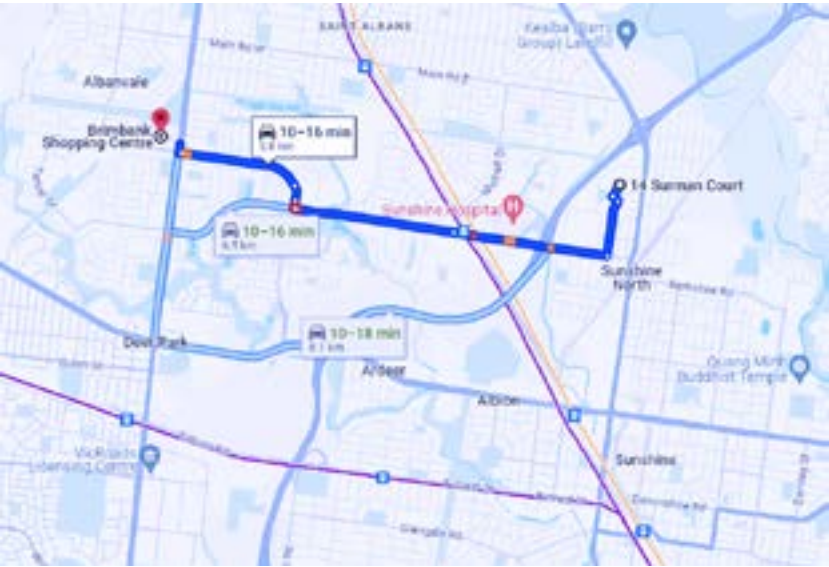
User 1: A housewife



Gia is a housewife, on a beautiful Tuesday afternoon she decided to go for grocery shopping at Bimbank Central Shopping Centre. She lives at 14 Surman Court, Sunshine North. She needs to decide whether to go by public transport or by her car.

Distance between two places: 5.80 km

Journey by Car

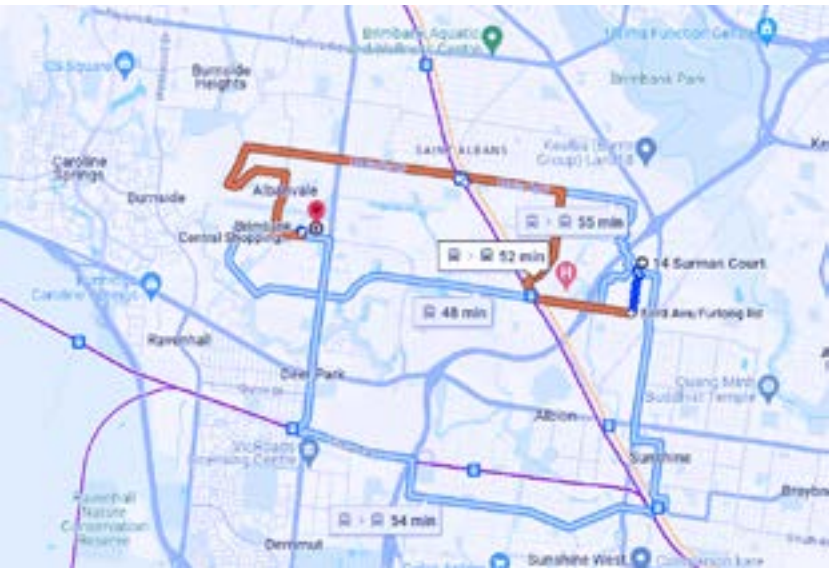


Journey by Car (Source: Google Maps 2024)



Total Journey Time:
10- 16 minutes

Journey by Public Transport



Journey by Public Transport (Source: Google Maps 2024)



Transport Option: Bus
Total Journey Time :
48-55 minutes

Option 1:

Route 423

Total travel time: 48 minutes

Walk to the bus stop: 32 minutes (3.2 km)

Total walk time: 33 minutes

Option 2:

Route 408 + Route 424

Total travel time: 52 minutes

Walk to the bus stop: 12 minutes (0.85 km)

User 2: A part-time employed young student.



Ronney is a part-time employed young student. He live at 36 Meadowbank Drive, Sunshine North and works at Kmart Keilor. He does not have a car. Every Saturday he travel to work by public transport.

Distance between two places: 5.12 km

Journey by Public Transport



Journey by Public Transport (Source: Google Maps 2024)



Transport Option: Bus
Total Journey Time :
41-52 minutes

- Option 1:
- Route 408 + Train (Sunbury) + Route 421
- Total travel time: 43 minutes
- Walk to the bus stop: 6 minutes (0.45 km)
- Total walk time: 14 minutes
- Option 2:
- Train (Sunbury) + Route 421
- Total travel time: 41 minutes
- Walk to the train station: 22 minutes (1.5 km)
- Total walk time: 26 minutes

3. OPPORTUNITY STATEMENT

There is an opportunity to establish a network that prioritizes active transport and micromobility.

The transport futures envisioned will contribute to shaping the amenity and development of Sunshine North’s economy.



Sunshine North Railway track (photo credit: R. Rafiq)

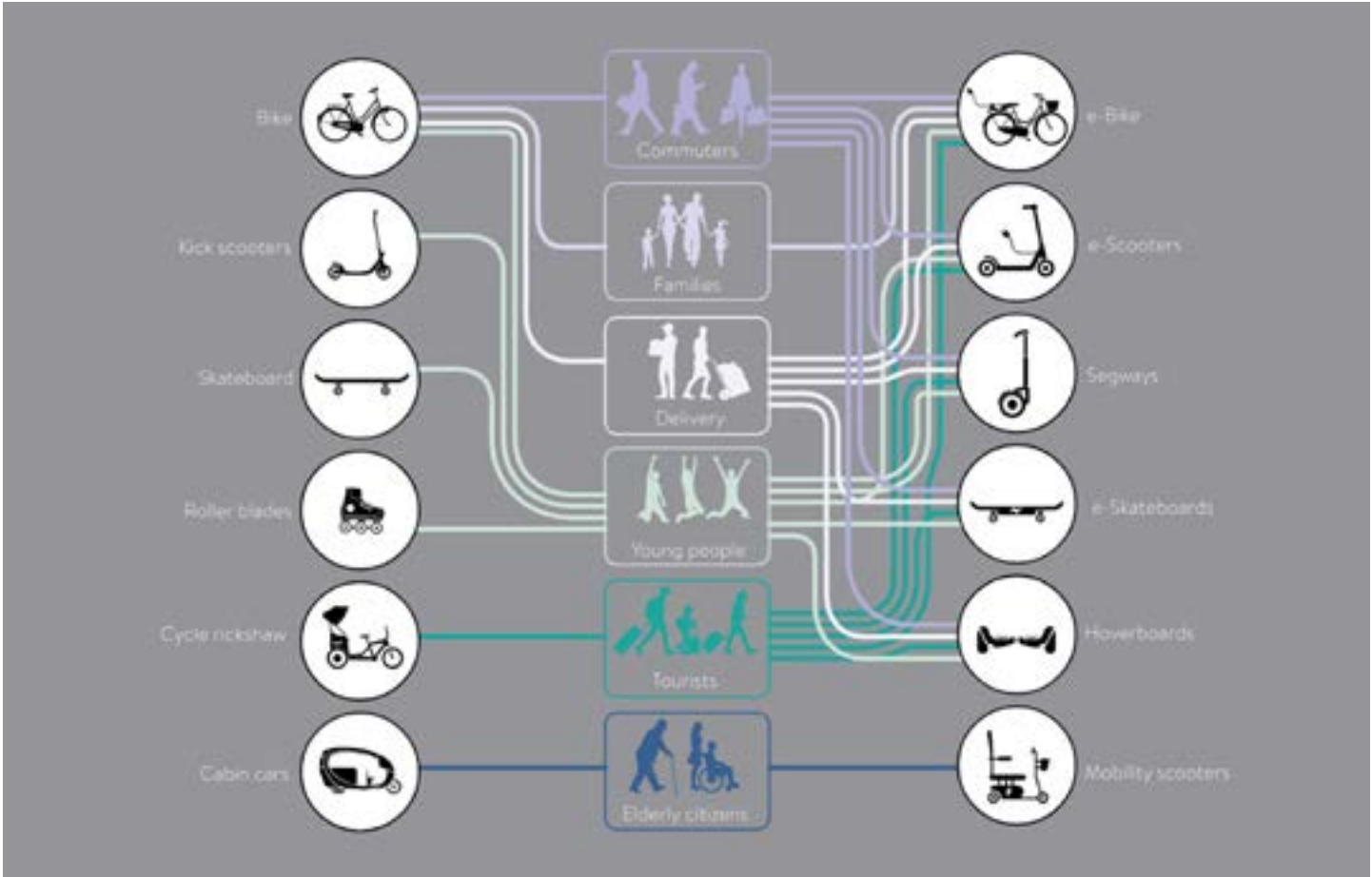
What is Micromobility?

Micromobility involves compact, lightweight modes of transportation that can accommodate up to three persons. Micromobility is primarily used for short trips within cities, typically up to 10 km. ¹

Micromobility vehicles are powered by electricity or human power and have a maximum speed limit of 25 km/h for safety. The word micromobility refers to both private and shared transportation.¹

With cities expected to house two-thirds of the world's population by 2050, it's crucial to address the issue of urban mobility.¹

Micromobility Users



Micromobility Users (image source: Buro Happold and Cassani, Micromobility Street Design Manual.)

1. Buro Happold and Cassani, “Micromobility Street Design Manual.”

4. SCOPING THE MOBILITY PROJECT



Sunshine North car dominance(image source: James Whitten)

Critical Uncertainties

1. Climate Change

Victoria is already experiencing:

- An increase in the frequency of extreme heat events;
- An increase in dangerous fire weather and the length of the bushfire season;
- A decrease in average rainfall, especially in cooler months.¹

Cities Central Role in Climate Change, 2019 (image source: Smart Cities World)



Climate change has the potential to impact all aspects of the Transport system’s operations, maintenance and construction which causes:

- Harm to assets;
- Disruption of transport services, operations, maintenance and constructions;
- Disrupted access;
- Discomfort to transport staff and users;
- Increased public safety hazards.¹

2. Geopolitics affecting industrial production

Geopolitics affects the nature of a countrues demands and will also then affect what a country produces:

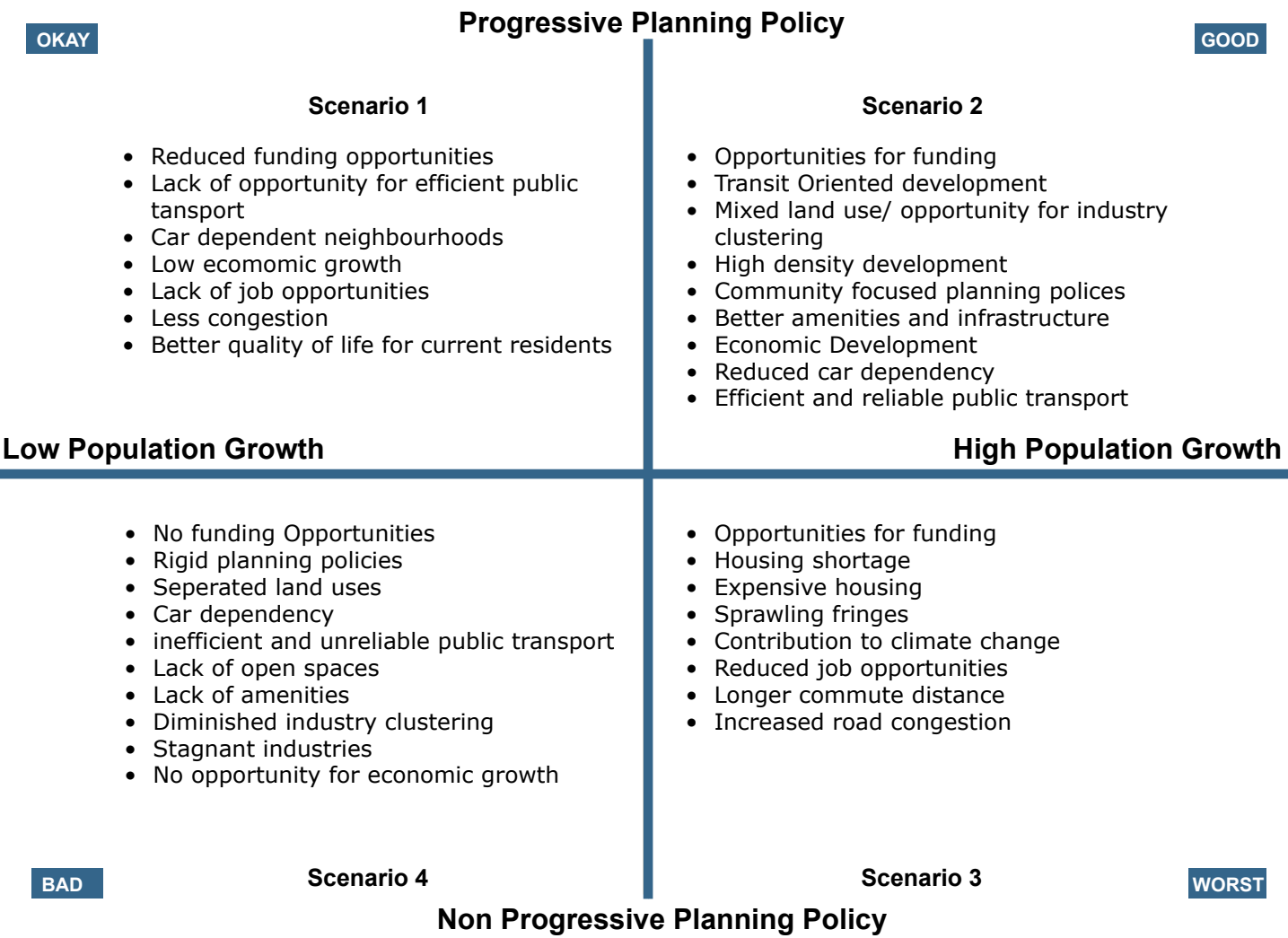
- Supply and demand;
- Changing political circumstances;
- Changing political ideologies;
- Changing global trade ;
- Various vested interest;
- Increased public safety hazards.



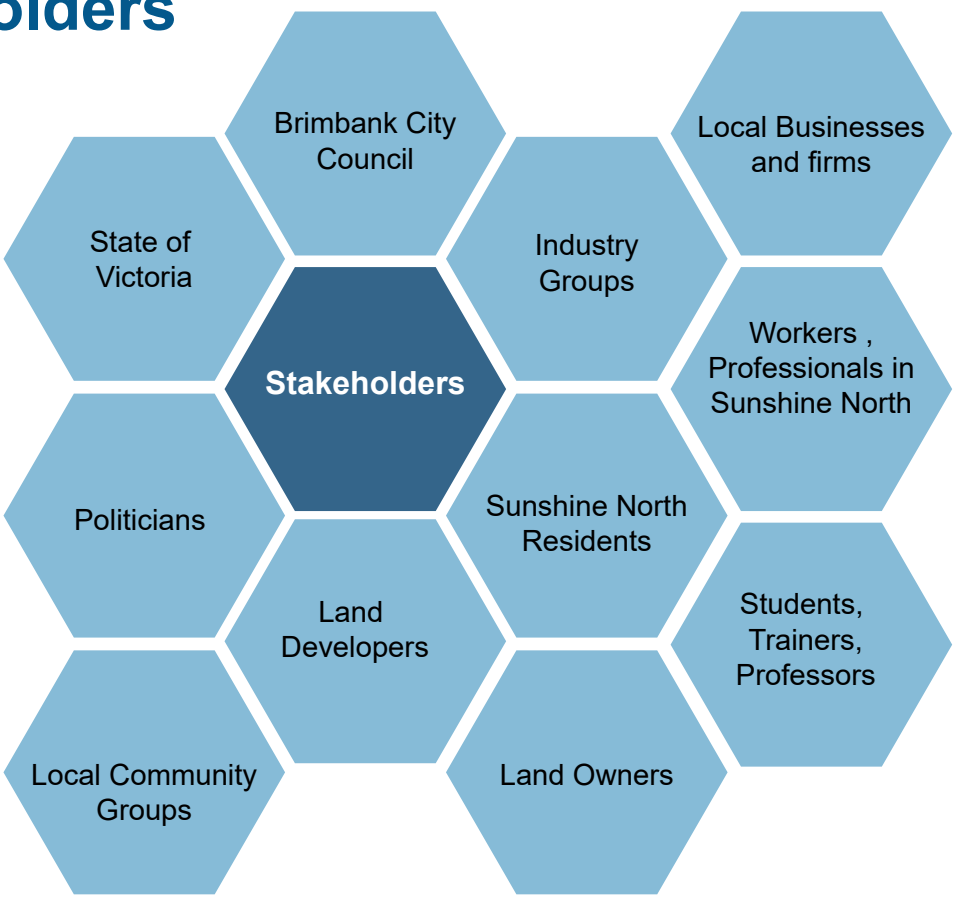
Geopolitics affecting industry, 2023 (image source: Loss prevention media)

1. Department of Environment, Land, Water and Planning, “Victoria’s Climate Science Report 2019.”

Scenario Matrix



Stakeholders



Policy and Regulatory Analysis

State Policies



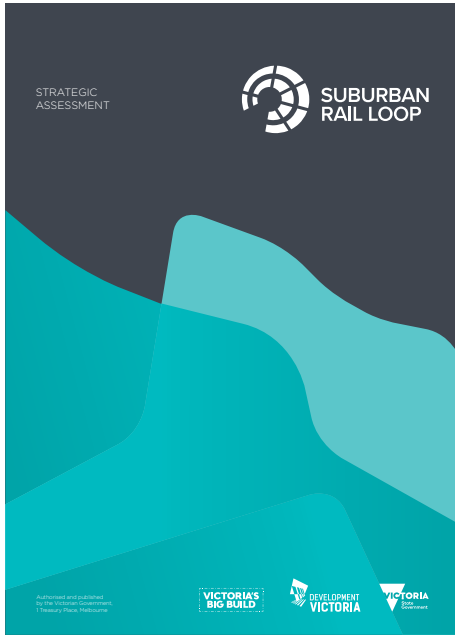
Plan Melbourne 2050, 2017 (image source: Department of transport and planning)



Sunshine NEIC (image source: Victorian Planning authority, State Government of Victoria)



Melbourne airport rail (image source: Victoria's big build, State Government of Victoria)



Suburban rail loop (image source: Victoria's big build, State Government of Victoria)

The “Sunshine National Employment and Innovation Cluster Draft Framework Plan” presents a 30-year goal for Sunshine to become a prominent suburban employment and business destination in Melbourne’s west. It focuses on increasing transportation connections, public amenities, and encouraging investment in healthcare, education, and business services. The plan covers strategic goals such as establishing place-based management organisations, offering integrated transport networks, constructing trail and cycling paths, and releasing land for job creation. It strives to promote innovation, livability, and environmental sustainability while also providing certainty for future development.

Local Government Policies



Brimbank Economic Development Startegy (image source: Brimbank City Council)



Sunshine Precinct Opportunity Statement 2021 (image source: Brimbank City Council)

Local level government policies are largely centered around the guiding state level ‘Plan Melbourne’. Although the guiding principles find their origin in the state level document there are a number of strategies and actions formulated and acted upon by the local government.

Precinct Opportunity Statement:

- develop sunshine as a major employment center in the west, that connects to the airport, metropolitan and regional centers
- leverage major rail investment to deliver private investment;
- attract new investment to create jobs in competitive industries;
- provide greater diversity of jobs and intensify employment activity
- deliver high amenity interface with residential areas, open space, and along significant streets

Brimbank Economic Development Strategy 2022-2027:

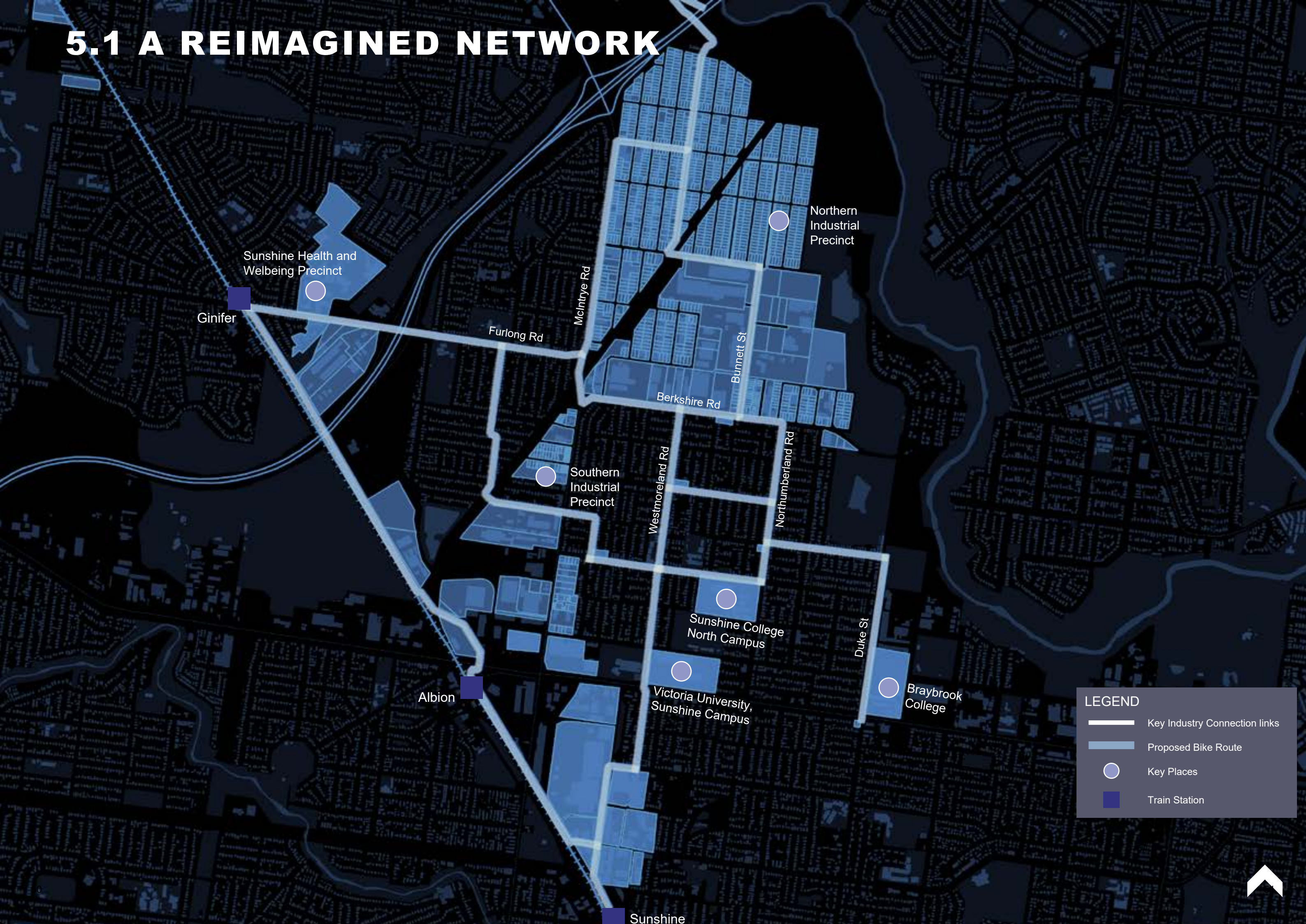
- 1) Facilitate Sunshine NEIC's development as a major provider of tertiary education, health-related training, healthcare services, medical research, education, transport, business and retail services
- 2) Encourage significant growth and land use change to leverage transport infrastructure investment and improved connectivity within and to the Sunshine NEIC
- 3) Leverage investments in transport-related projects by supporting accommodation, entertainment and conference facility uses in the Sunshine Metropolitan Activity Centre
- 4) Improve connectivity to the Health, Wellbeing and Education Precinct, Sunshine and St Albans Activity Centres and industrial precincts and,
- 5) Develop the Sunshine NEIC to provide major public service facilities for the growth areas of Melton, Sunbury, Wyndham and beyond to the western region of Victoria.

5. PROJECT PROPOSITION



Sunshine North infrastructure(image source: James Whitten)

5.1 A REIMAGINED NETWORK



LEGEND

Key Industry Connection links

Proposed Bike Route

Key Places

Train Station

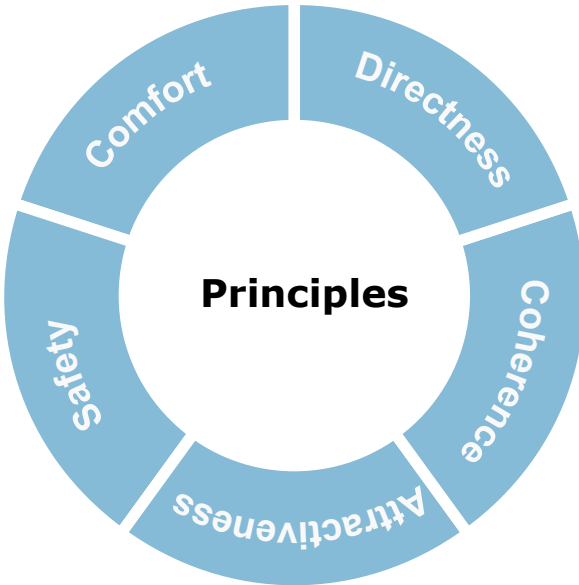


Pillars of Micromobility



Pillars of micromobility (image source: Buro Happold and Cassani, Micromobility Street Design Manual.)

Planning Goals and Principles



Theme 1 Health and Safety

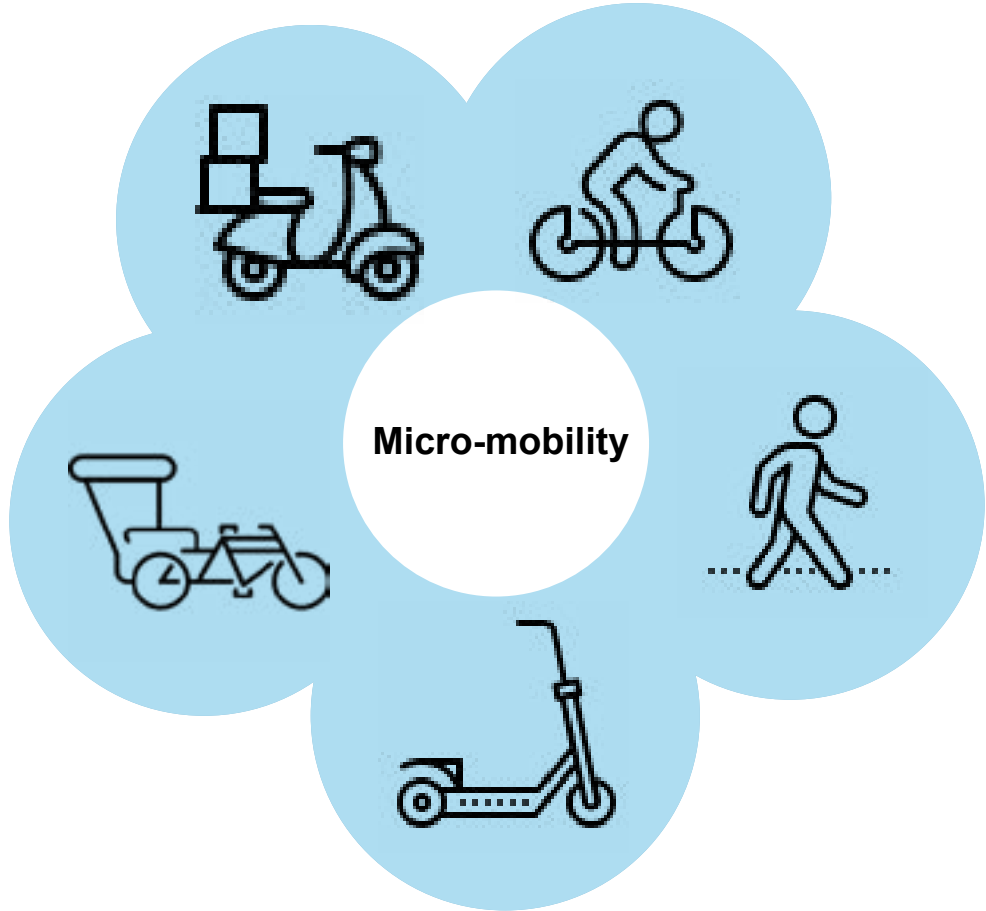
Startegy Goals

- 50% of Sunshine North residents out of their cars by 2051.
- Significant reduction in automobile incidents within the network.
- Implement an equitable network that facilitates the uptake in usage by all.

1. Equitable, Accessible and secure Transport for all.

Action

- 1.1 Redesign streets to facilitate active transport usage.
- 1.2 Promoting the modal shift from motorised to non-motorised vehicles by implementing cohesive network.
- 1.3 Advocate awareness and acceptance of active transport as a mode of transport.
- 1.4 Reviewing speed zones, Designing streets for all.
- 1.5 Designing shared streets.



Strategy Goals

- Networks that restructure the industry base.
- Network of industrial clusters that don't rely on cars by implementing an equitable network that facilitates the uptake in usage by all.
- Supporting Future made in Australia initiative: Inviting emergent industries into the locale: pharmacy industry

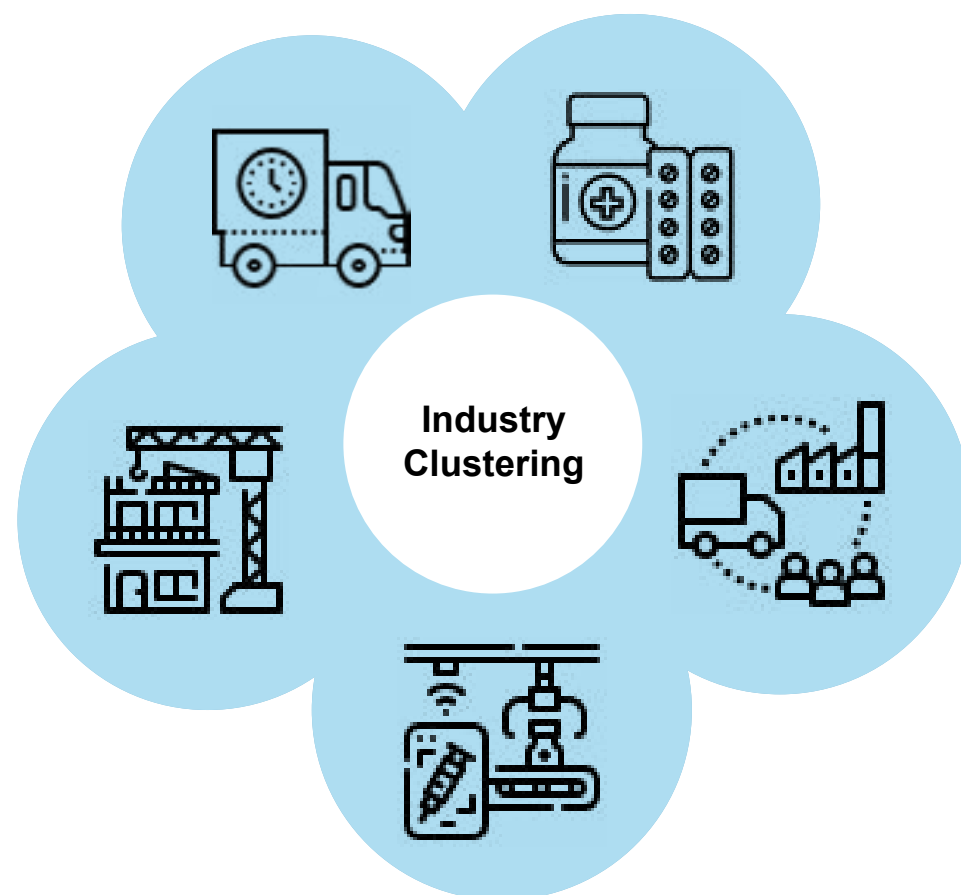
2. Supporting growth through smarter planning

Action

2.1 Planning reappraisal to induce an industry response to the locale

2.2 Encourage target industries and firms into the precinct with promotions and incentives.

2.3 Shaping the urban form to be attractive to desirable and attractive firms - i.e., safe passages of travel, high levels of amenity and urban form that is low-emission.

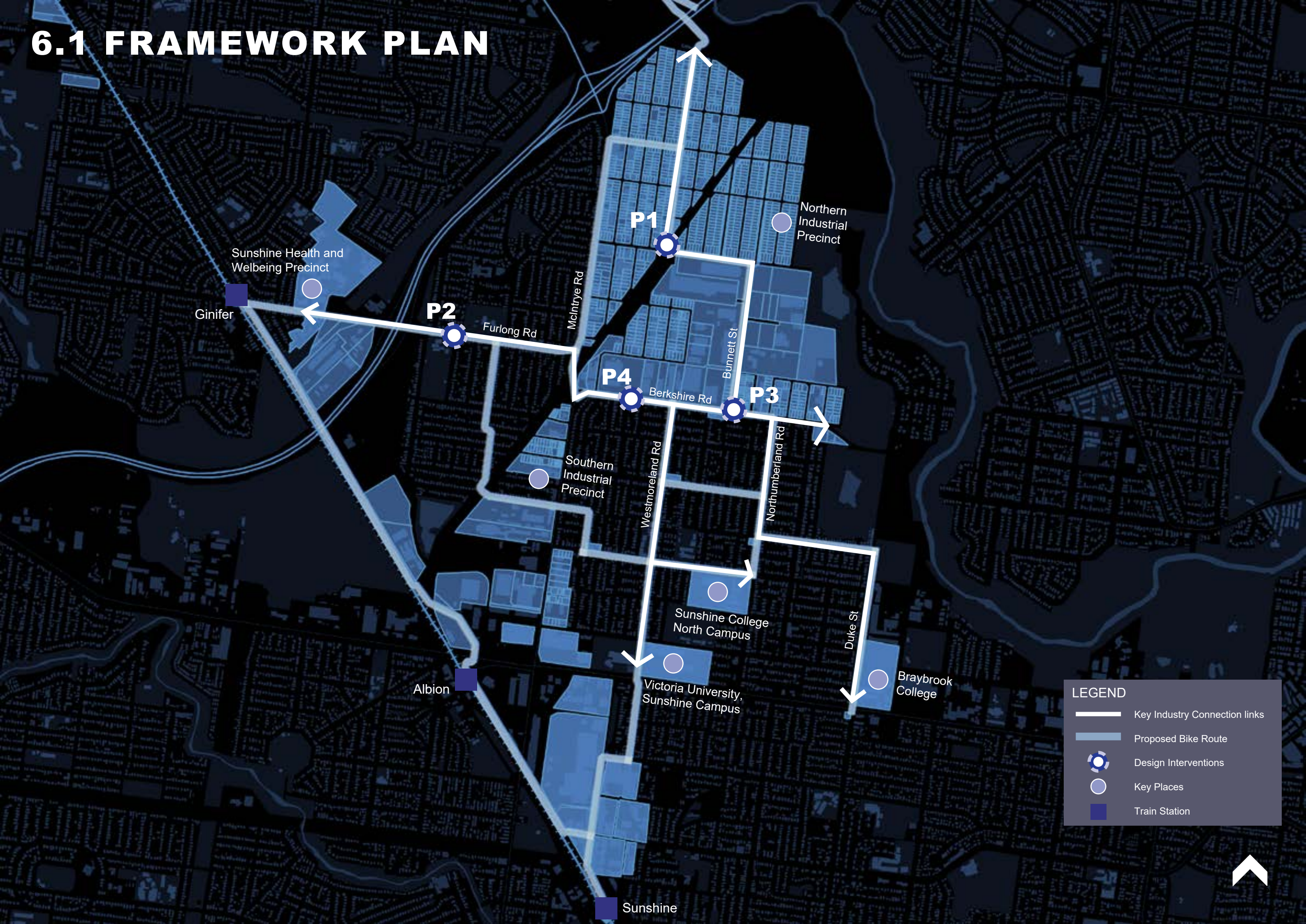


6. DESIGN INTERVENTIONS



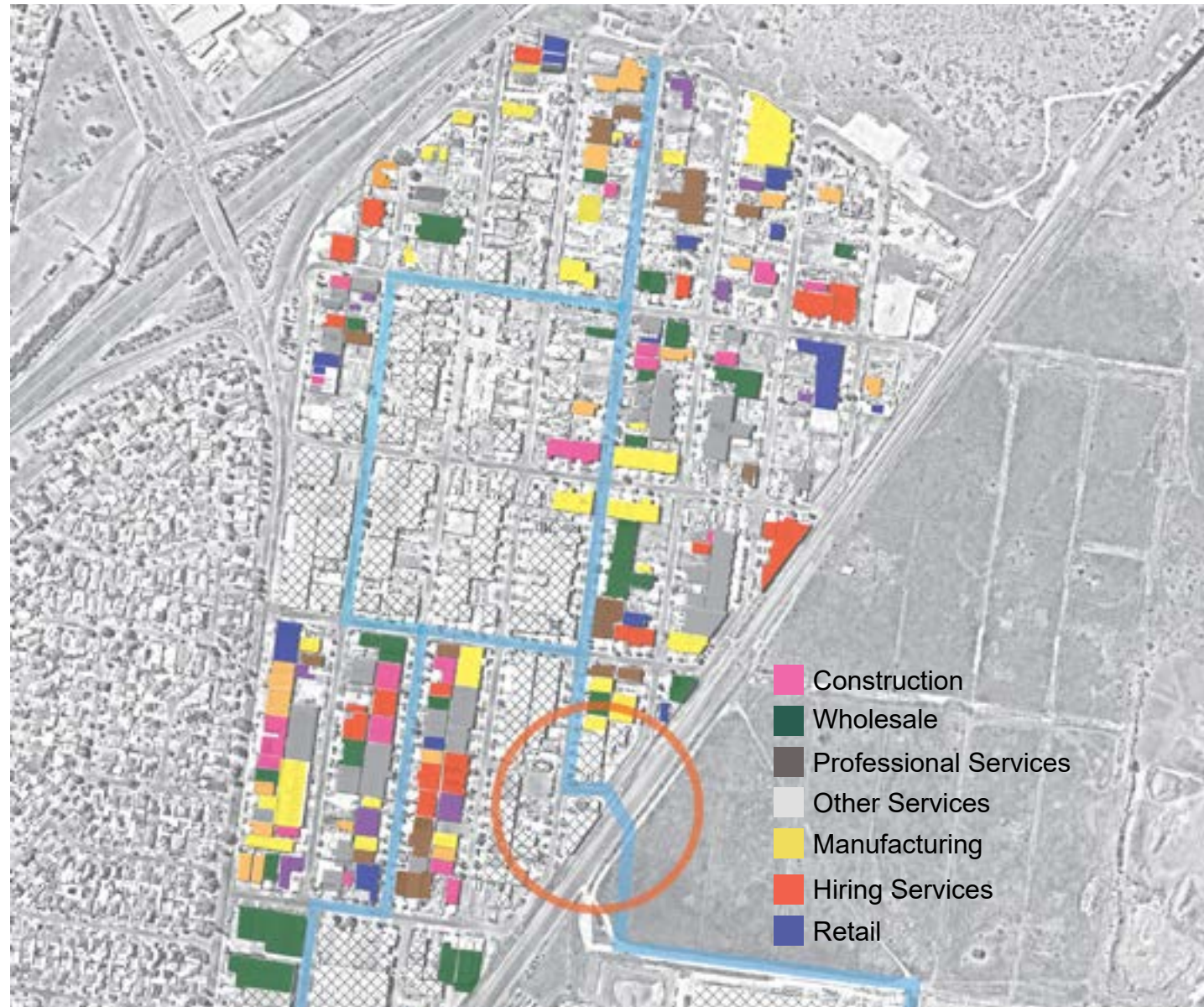
Sunshine North Open Space(image source: James Whitten)

6.1 FRAMEWORK PLAN



Bridging the gap

Connecting industries by micro-mobility bridge.



The diverse mix of industries in the northern industrial precinct warrant a connector. We envision a micromobility bridge that will be utilised by a plethora of electric vehicles delivering items, picking up parts, commuting and the like. We hope that the implementation of the micromobility bridge will not only be utilised by current firms within the locale but also be a symbol to potential and aspiring firms that the Sunshine North industrial precinct is an precinct of opportunity.

The location of the bridge is largely guided by the strategy of located a distribution center in the currently highly contested 'Soloman Heights area' (see next page). This area has been and is currently going under utilised due to various reasons. Strategies

Sunshine North Industries 2024(image source: R. Rafiq)



Micromobility Bridge Location, 2024(image source: R. Rafiq)

guiding the solving of the Soloman Heights area will be critical in unlocking the potential for the Sunshine North industrial precinct to flourish as a precinct embracing the future of industrial mobility.



Micromobility Bridge 2024(image source: R. Rafiq)

P2 Re-imagined Furlong Road



Road section along Furlong road (image source: R. Rafiq)



Furlong Road is the major arterial that connects Sunshine North to the SHWEP and Ginifer Station.

Currently the road is allocated by the Department of Transport and Planning as being part of the strategic cycling corridor but the experience of the road lacks service. There is a no demarcations or infrastructure protecting potential alternative modes of transport.

Re-imagined Furlong Road (image source: R. Rafiq)

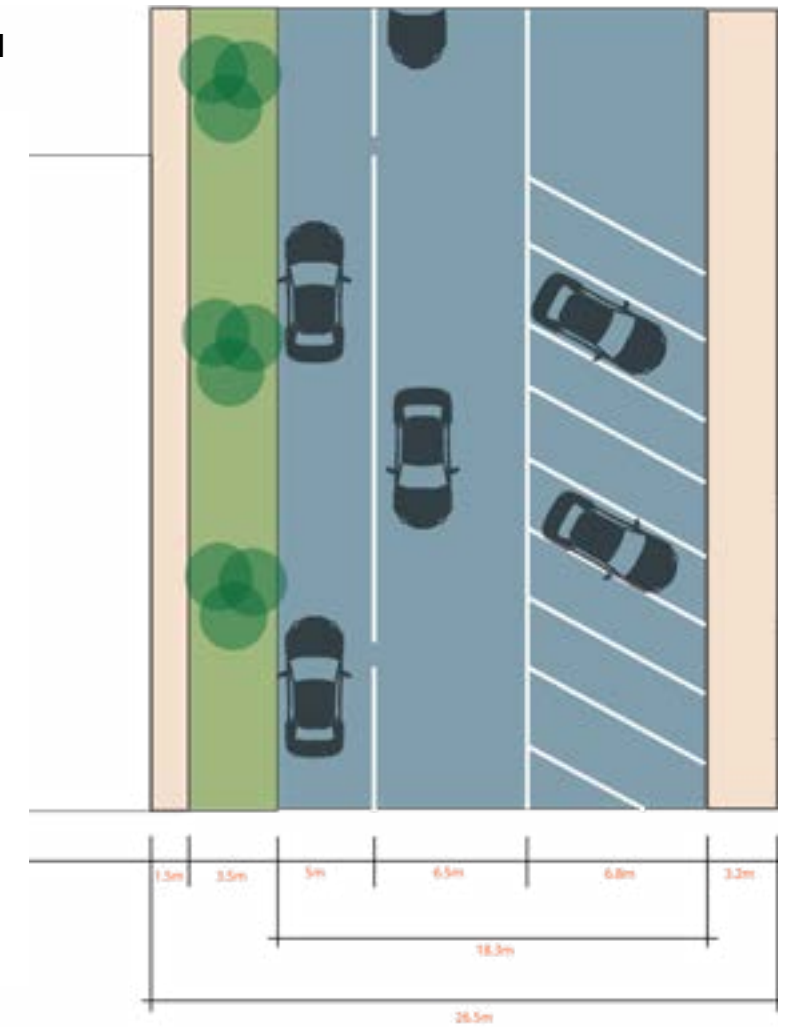


Re-imagined Furlong Road (image source: R. Rafiq)

Current Conditions: Furlong Road

- The current conditions on Furlong Road adjacent to a strip of services and retail embodies no infrastructure for non-car usage.

Road section along Furlong road (image source: R. Rafiq)

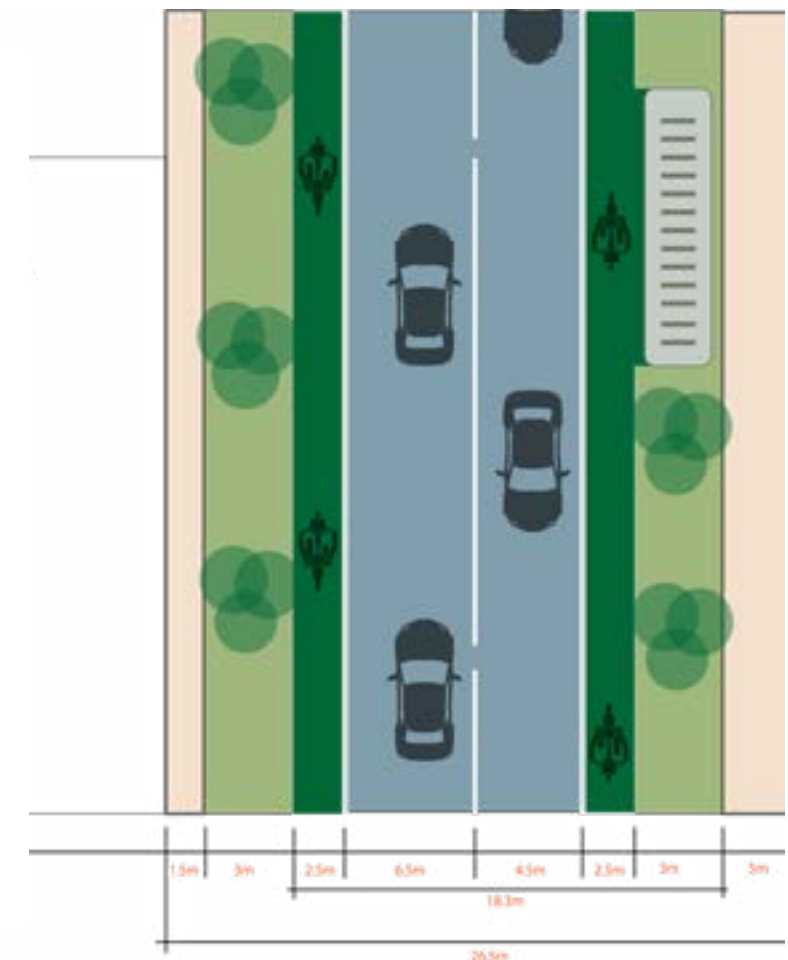


Re-imagined Conditions: Furlong Road

- By removing the parking and reallocating it to the rear of the strip mall, there is a surplus of road space to be utilised for a more equitable level of service.

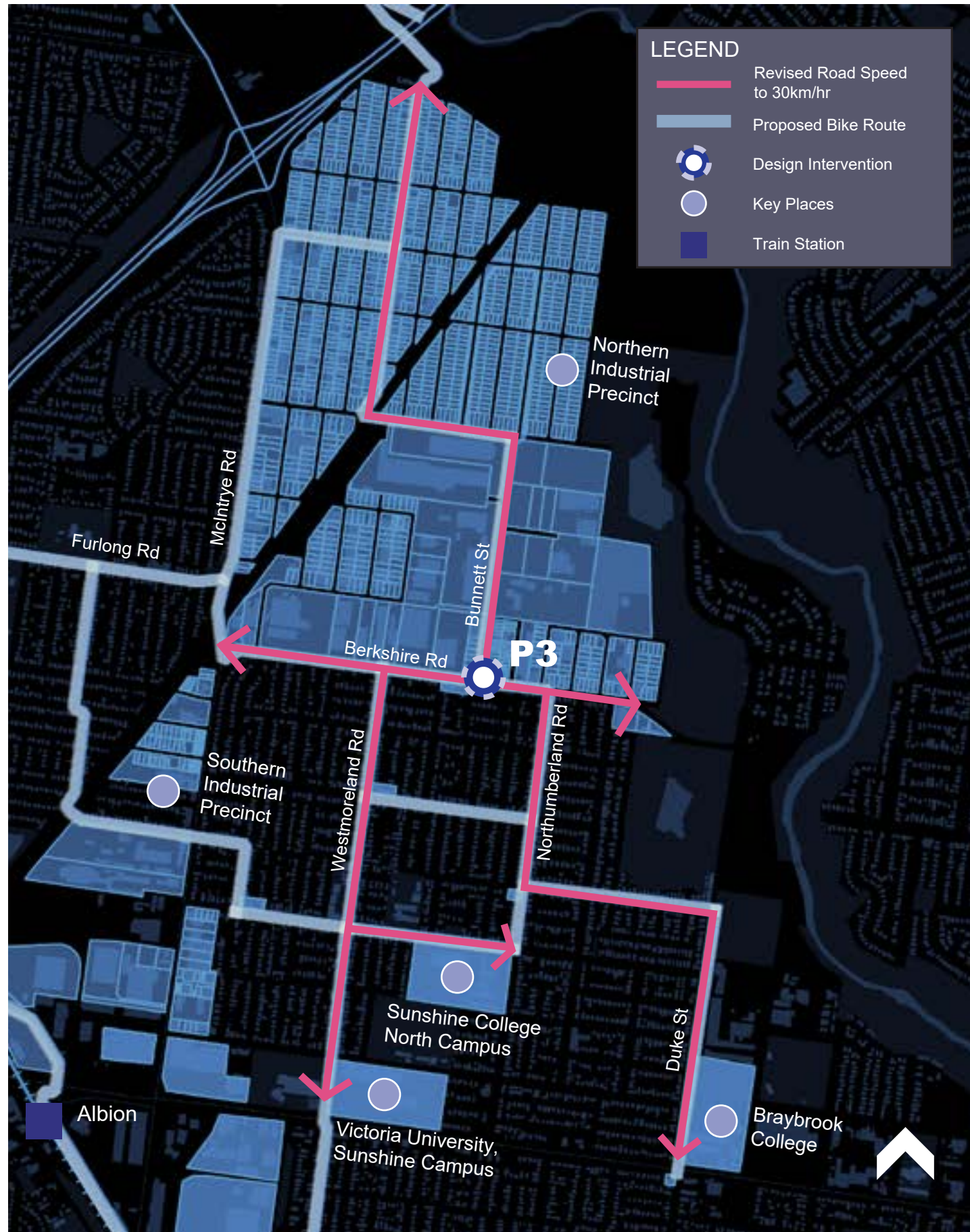
Introducing cycle lanes on either side with a dedicated bicycle park on the retail side will ensure the safe passage of commuters.

Road section along Furlong road (image source: R. Rafiq)



Re-imagined Streetscapes

Redesigned Streets and Revised Road Speeds



LEGEND

- Revised Road Speed to 30km/hr
- Proposed Bike Route
- ⚙️ Design Intervention
- Key Places
- Train Station

Design Intervention Map (image source: Anagha Khude)

Case Study

1. Avenue des Champs-Élysées Paris, France

- Commercial Street
- Two-way Boulevard
- A high touristic attraction
- Located in the heart of the city ¹



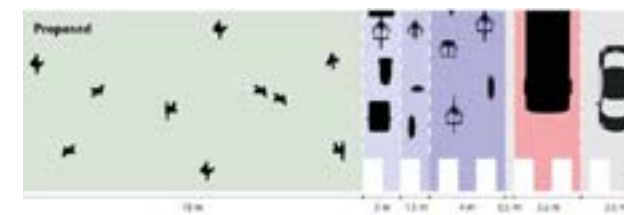
Avenue des Champs-Élysées, Paris(image source: Buro Happold and Cassani, Micromobility Street Design Manual.)



Existing Plan (image source: Buro Happold and Cassani, Micromobility Street Design Manual.)



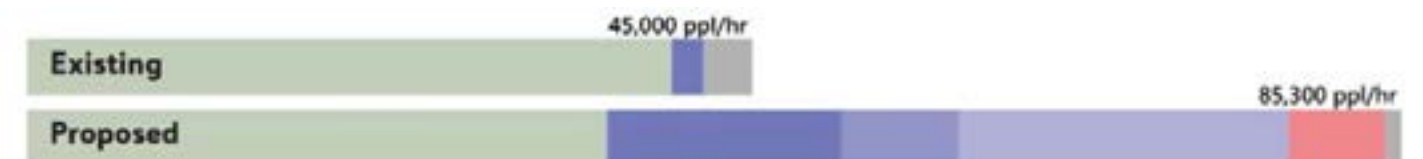
Existing Scenario (image source: Buro Happold and Cassani, Micromobility Street Design Manual.)



Proposed Plan (image source: Buro Happold and Cassani, Micromobility Street Design Manual.)



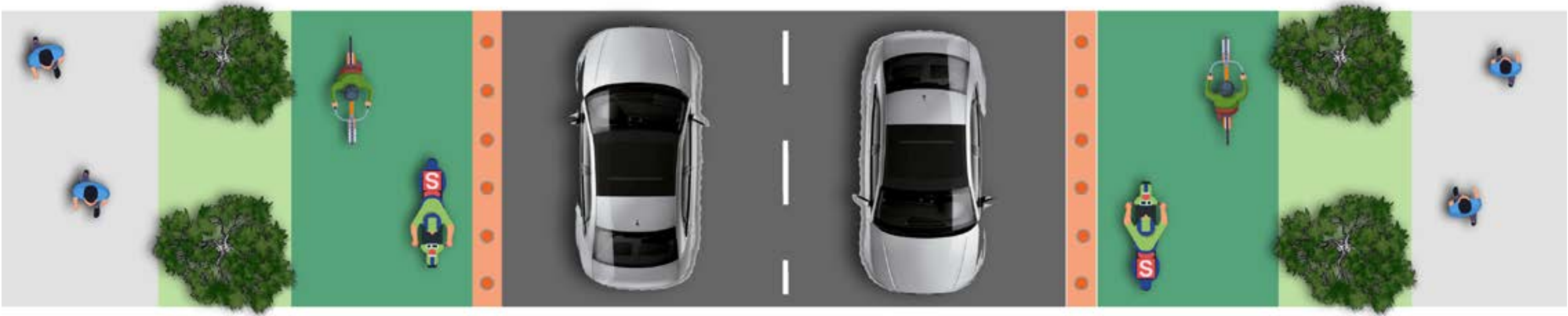
Proposed Vision (image source: Buro Happold and Cassani, Micromobility Street Design Manual.)



Expected Footfall (image source: Buro Happold and Cassani, Micromobility Street Design Manual.)

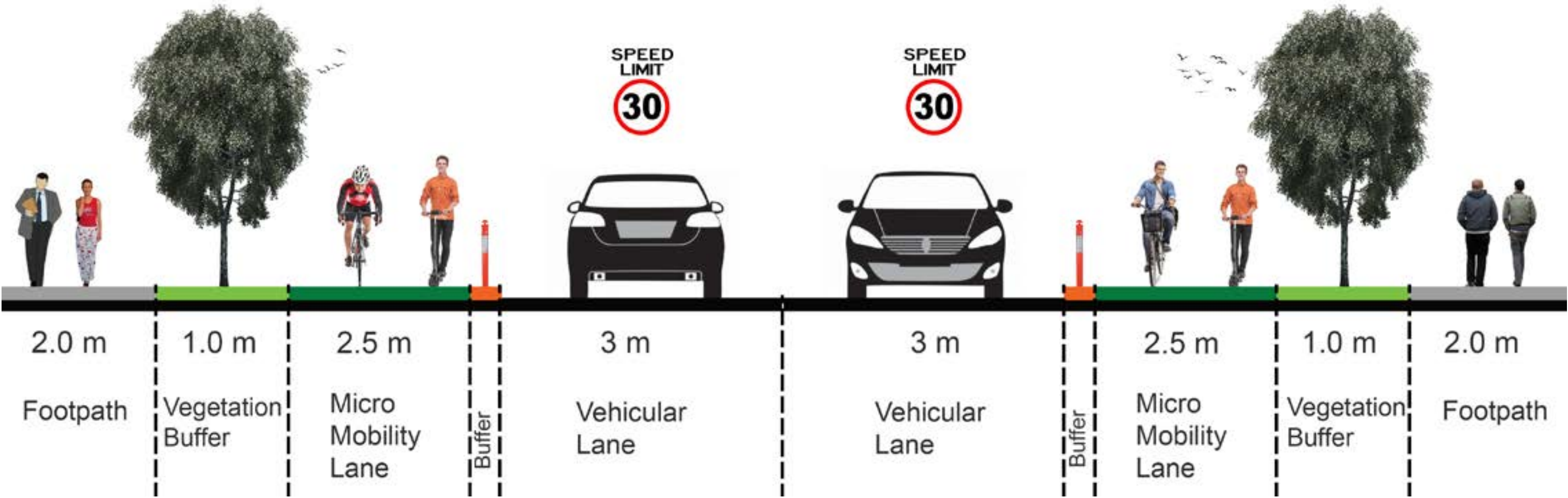
1. Buro Happold and Cassani, “Micromobility Street Design Manual.”

Proposed Design of Streets



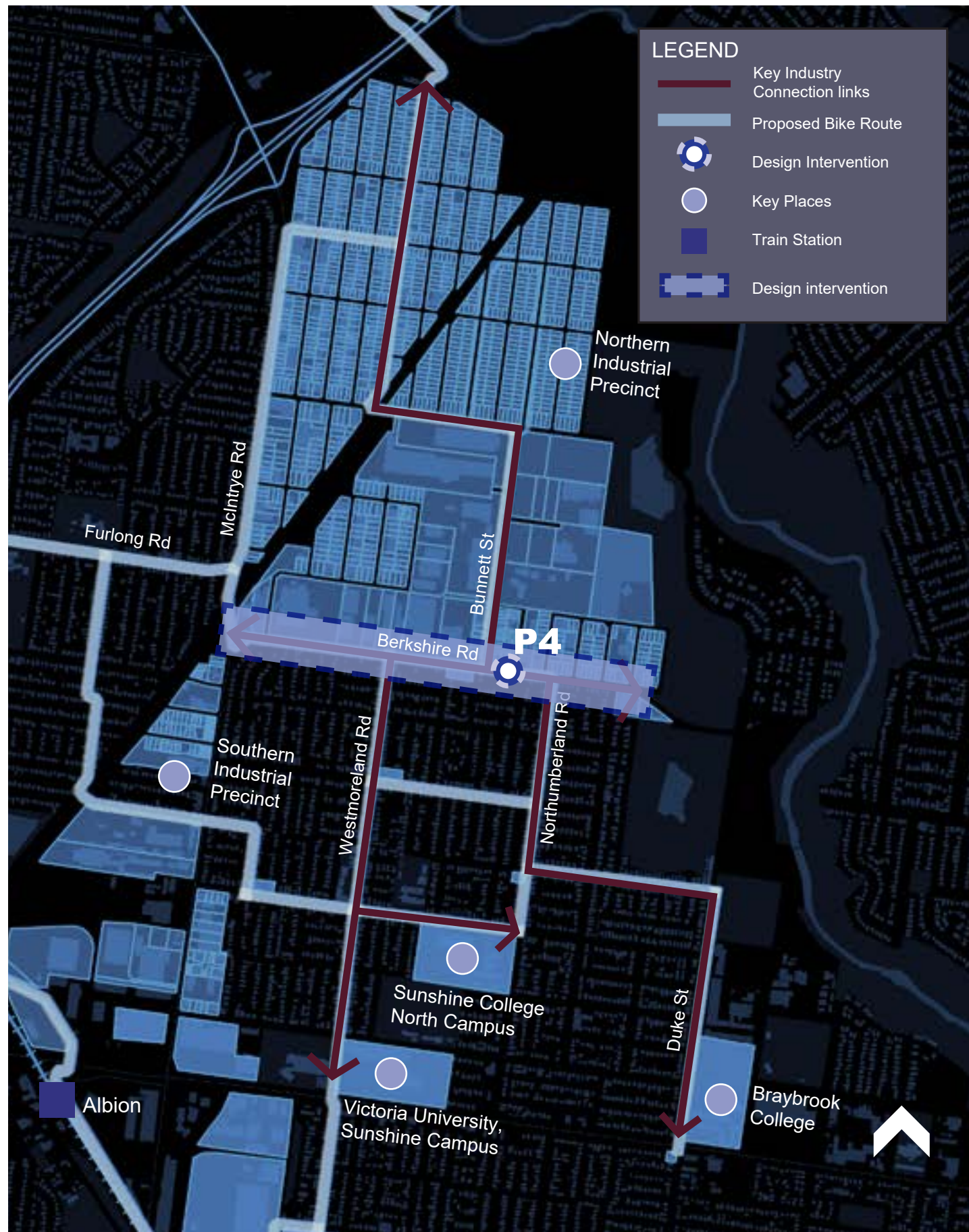
- Low vehicle speeds will result in a safer traffic flow because all cars on the same surface will be regulated at the same pace.

Plan (NTS)
Road Plan (image source: Anagha Khude)



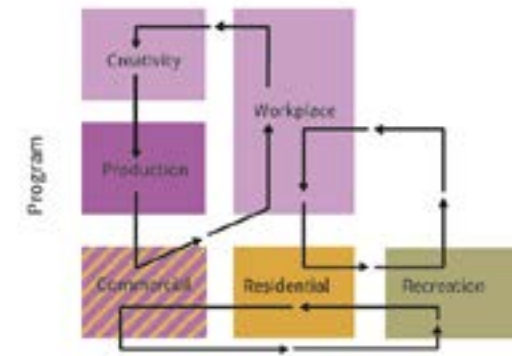
- This proposed design will create opportunities for budding microbusinesses to assist economically energise a new region in Sunshine North.
- This design will allow for the intervention of green areas and public breakout spaces, as well as the reactivation of underperforming retail areas.

Section (NTS)
Road Section (image source: Anagha Khude)



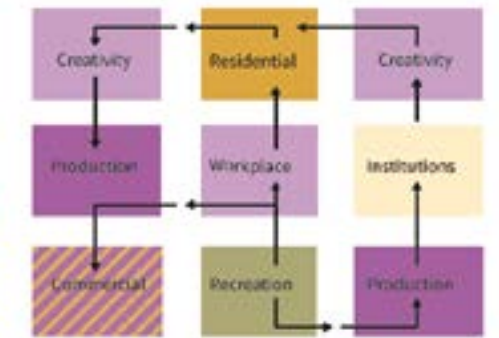
Design Intervention Map (image source: Anagha Khude)

Segregated Areas Network

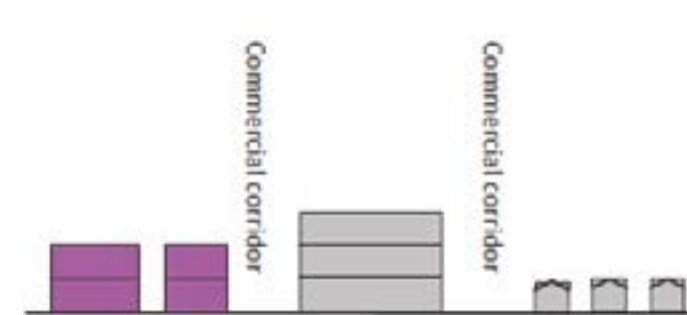


Segregated areas Network (image source: Hatuka and Ben-Joseph, New Industrial Urbanism: Designing Places for Production)

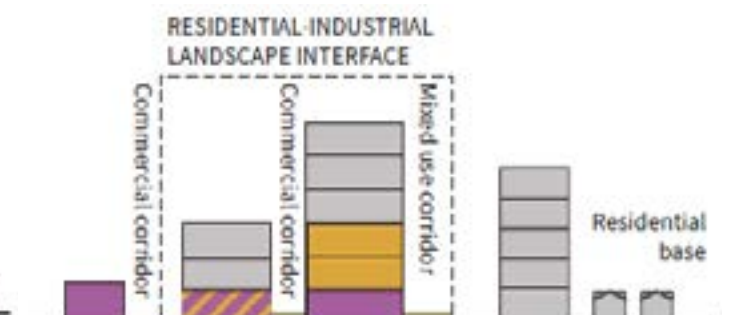
Consolidated Areas Network



Consolidated areas Network (image source: Hatuka and Ben-Joseph, New Industrial Urbanism: Designing Places for Production)



Separation Industrial-Residential Nexus (image source: Hatuka and Ben-Joseph, New Industrial Urbanism: Designing Places for Production)



Transition Industrial-Residential Nexus (image source: Hatuka and Ben-Joseph, New Industrial Urbanism: Designing Places for Production)

Precedent

Towards an Integrated System: Eastern Market Neighborhood, Detroit

The Eastern Market Neighbourhood plan in Detroit, United States, is a recent example of combining industrial services with other uses, including housing. The new plan for the area sought to establish a vision for development that would chart an implementation strategy for the growth and integration of food-business-centered production and distribution, housing, and the use of unoccupied land, all while including ecological and sustainable practices.¹

Mixed uses in industrial regions improve the likelihood of public spaces becoming dynamic, lively locations where public spaces are used by a broader segment of the community, in addition to local residents and users. Increased exercise may result in more incorporation into social, cultural, and educational programmes, as well as increased career opportunities.¹

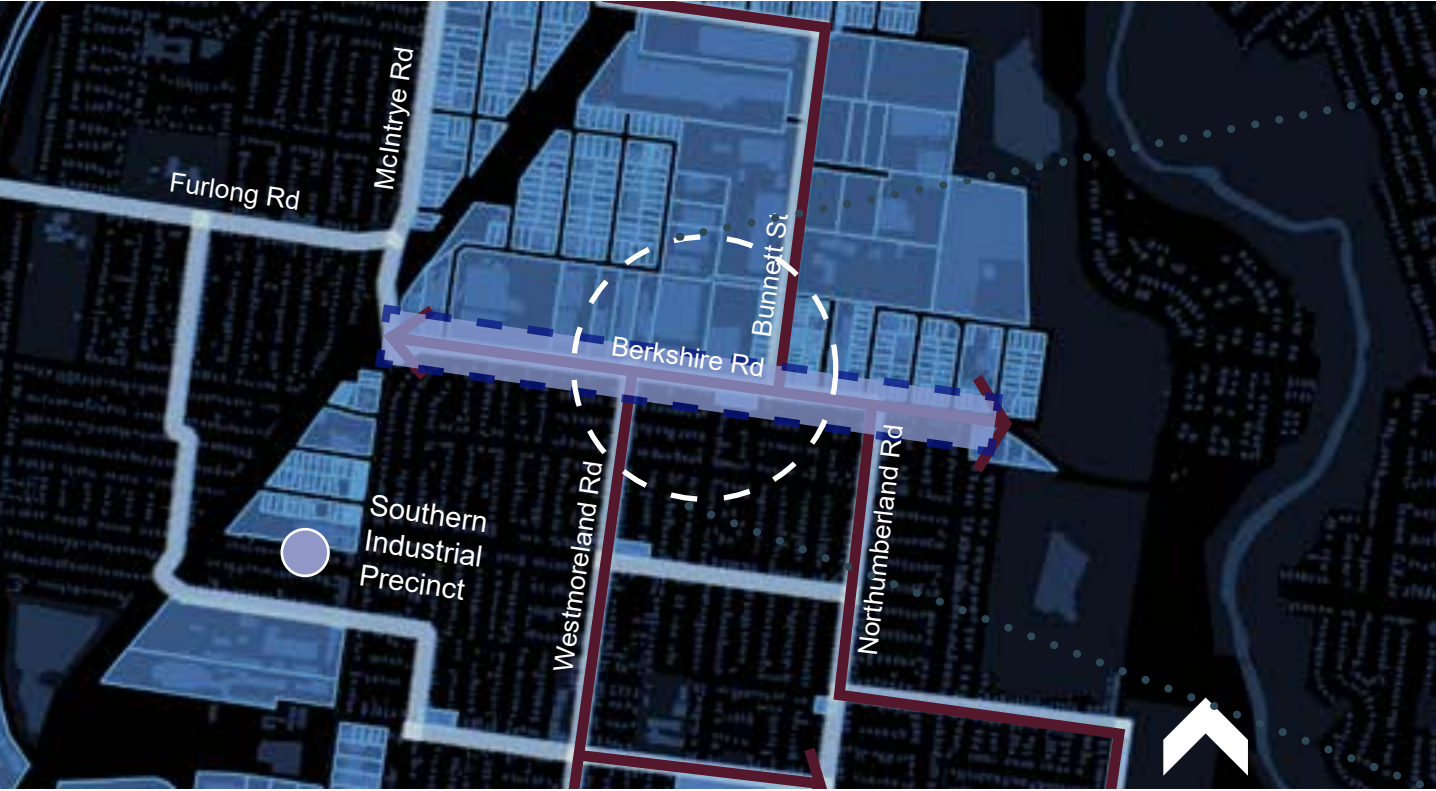


Eastern Market Plan, Detroit, MI, USA. Maps and drawings courtesy of city of Detroit, Utile Inc. and Michael Van Valkenburgh Associates.

Eastern Market Plan, Detroit (image source: Hatuka and Ben-Joseph, New Industrial Urbanism: Designing Places for Production)

1. Tali Hatuka and Eran Ben-Joseph, "New Industrial Urbanism: Designing Places for Production", 1st Edition (Routledge, 2022).

Proposed Integration of Land Use Zones along Berkshire Rd

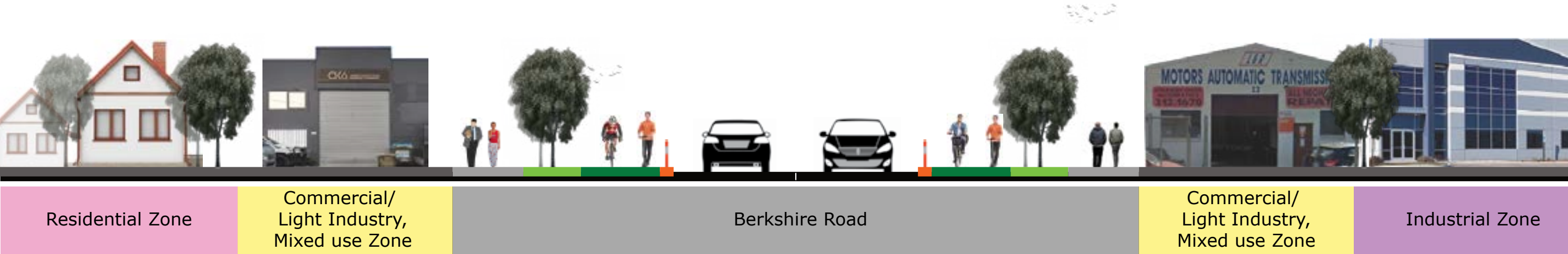
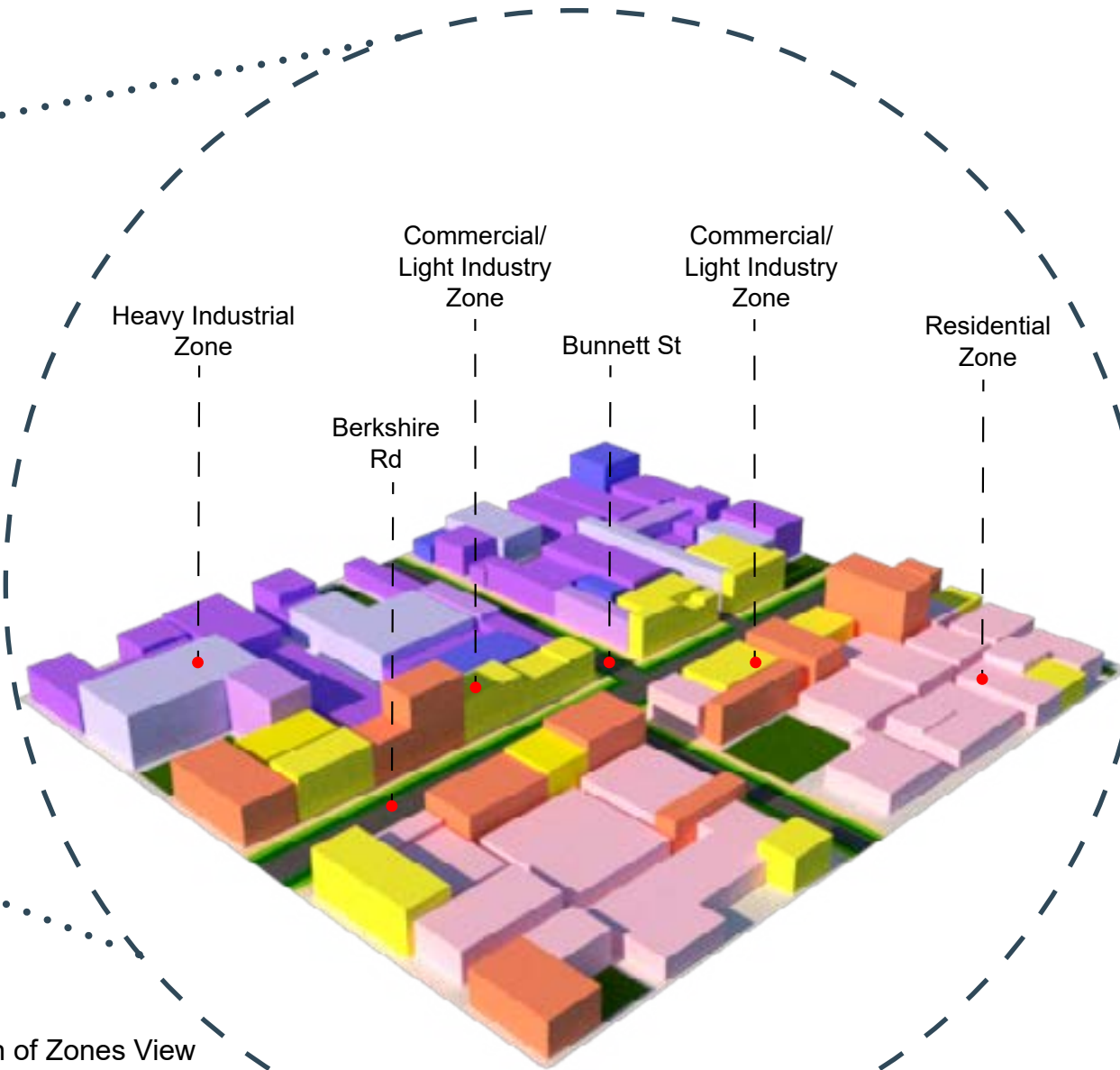


Location Map (image source: Anagha Khude)

- Implementation of regulatory revisions.
- Land use revision to change from heavy industry to light industry and commercial zones.
- Streetscape improvements supporting micro mobility networks.
- Integration promotes hybrid urban development, incorporating live/work and/or fully residential buildings.¹

Integration of Zones View

A View of Integration of Zones (image source: Anagha Khude)



Section (NTS)

A section through Berkshire Road (image source: Anagha Khude)

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